

Guest Lecture

Dr. William E. McCarthy

Object Oriented Business Information Systems

By Kip M. Twitchell

Global Subject Matter Expert

IBM

April 15, 2020



- Introduction
- Transactions vs. Balances in Analytical Processes
- The Daily Financial Cycle
- Minimum Cost for Maintenance of Financial Data
- Sample Application Technologies:
 - IBM's Scalable Architecture for Financial Reporting (SAFR)
 - Svayam Infoware's Financial Period End Management (FPEM) Application
- Sharealedger.org

Kip Twitchell – Global Subject Matter Expert, Financial Systems



Overview



Kip Twitchell is a Global Subject Matter Expert with IBM Global Business Services (GBS) Financial Services Sector, with extensive experience as a consultant in financial management and business intelligence systems.

He has consulted on or led construction of systems enterprise wide and multi-national financial systems for 24 of the top 50 banks (by 2015 IFRS assets) and numerous Fortune 500 companies, including systems based upon Oracle, SAP and custom developed products.

He is CPA and expert in blockchain, big data analytics, ledger processes, and the field of business events-based reporting and the author of “Balancing Act: A Practical Approach to Business Event Based Insights” (2011), a monograph “Reinventing the Data Supply Chain: From Search to Metric Engines for Business,” (2015) and publishes a weekly vlog, “Conversations with Kip” and a blog at financialsystemseducation.com

Highlighted Capabilities

- Consolidated Data Supply Chains
- Metrics, analytics, and reporting
- Risk Data
- Cost Allocations
- Funds Transfer Pricing
- Ledger Processes
- Budgeting, Planning and Forecasting
- Financial Reporting
- Management Reporting
- Global Program Transformation
- Event Based Accounting

Functional Experience

- Finance
- Risk
- Data
- Analytics
- Financial Services
- Thought Leadership
- Solution Manager
- Program / Project Management

Education

- Bachelors and Masters in Accountancy, Brigham Young University, and
- Certified Public Accountant
- Languages: English, Japanese

Select Experience

Blockchain and Big Data R&D: Research and Development investigating the most recent technologies for suitability to finance systems, including Blockchain, Spark, AI, cloud based systems, Scala, Dataframes, semantic databases and others; building proof of concept next generation finance systems; defining a generalized framework for a consolidated data supply chain

Global Finance and Risk System Data Expert: Consulted with numerous Global Systemically Important Financial Institutions (GSIFs) on the principles of a Consolidated Data Supply Chain, enabling higher quality data for Finance, Risk, MIS and even CRM and other applications.

US Regional Financial System Road Map and Blueprint Phase of SAP Implementation: Solution Architect and Project Executive through financial system road map project, establishing a 5 year plan for financial systems replacement, including an approach to risk integration, and initial blueprint phases of SAP implementation, including Bank Analyzer

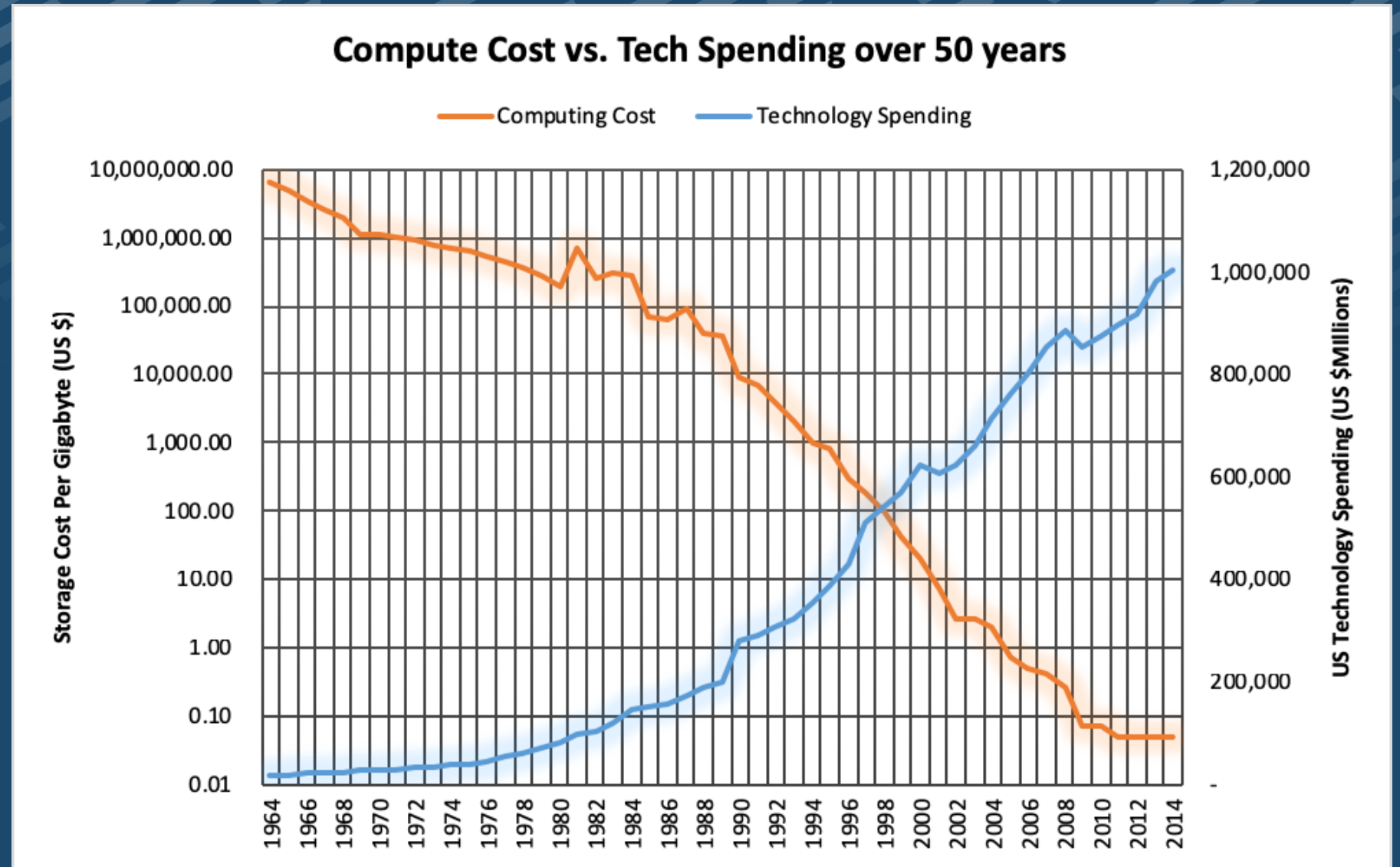
US Based Global Investment Bank, Finance IT System Road Map: Lead partner in development of a strategic road map of Finance Systems, including analysis of key requirements, existing assets, expected technology and functional developments, development and articulation and use of an industry standard finance system architectural framework, high level estimates to complete, and plan for execution

UK Based Global Bank, Finance Transformation Program: Project and architectural adviser for global role-out of platform providing a global general ledger, regulatory and management reporting environment, using accounting rules to post arrangement level detail directly from source systems and produce a daily trial balance in multiple currencies.. Project manager and solution architect responsible for organizing and overseeing the project team consists of over 400 staff, consultants and contractors both on-shore and off-shore in US, and adviser on approach and implementation in Hong Kong and UK

General Ledger and Financial Data Warehouse Implementations: System architect for enterprise wide statistical and financial data warehouse. Statistical data warehouse contains policy, and loss event level data. Financial data warehouse contains general ledger entries extracted from PeopleSoft General Ledger system, and allocated expense data. Architected statistical system parallel processing, source system extract, load and retrieval processing, and data storage. Architect for financial data warehouse and assisted in construction of code transformation, PeopleSoft load, extract, reconciliation, and reporting processes. Architected expense allocation process which mimics PeopleSoft allocation process but performs multiple levels of allocation at one time, allowing high data volume allocation to policy level detail.

A Proposed Approach to Minimum Costs for Financial System Data Maintenance

A White Paper by Kip M. Twitchell, Published at www.ledgerlearning.com



Primacy: Transactions—Efficiency: Balances

Dr. McCarthy's original REA paper makes the case that the primary data for all financial or analytical reporting is transactional data.

Transactions:

- The primary record of events
- Enables any analytical output from any combination of these attributes
- Eliminates redundant data and reconciliation

Yet the world is full of ledgers, consistently producing balances. Balances are efficient for computer processes and analytical processes.

Balances:

- Eliminate reaggregating historical transactions to state current positions.
- Almost all analytical processes begin by analyzing a balance

Customer Contract ID	Account Open Date	Interest Rate	Domicile	Obligation ID	Legal Entity	Cost Center	GL Account	Tran Code	Transaction Date	Source System ID	Currency	Amount	Description
7832485732	10/20/2000	10.00%	Phoenix, AZ	92432	45802A	30802381	1083100001	Payment	Sept. 14	Mortgage	USD	(782.00)	Monthly Mortgage Payment
4576483024	3/1/1991	2.07%	Mexico City	12312	45802A	30802381	1083100001	Payment	Sept. 20	Mortgage	USD	1,340.00	Monthly Mortgage Payment
9809249098	7/17/2012	0.28%	Madrid, Spain	23421	45802A	30802381	1083100001	Payment	Sept. 21	Mortgage	USD	(904.00)	Monthly Mortgage Payment
5893589357	7/4/2001	0.08%	Mexico City	23422	45802A	47683784	1083100001	Deposit	Sept. 22	TDA	USD	(299.00)	Savings Deposit
8324896847	3/22/1999	0.06%	Dallas Texas	34095	45802A	47683784	1083100001	Deposit	Sept. 22	TDA	USD	(67.00)	Savings Deposit
7832485732	10/20/2000	10.00%	Phoenix, AZ	92432	45802A	30802381	1083100001	Fee Rervasal	Sept. 25	Escrow Service	USD	(22.31)	Mortgage Escrow Fees
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Balances have downsides. Balances

- Are effectively a duplication of selected attributes and accumulated amounts of the transaction data. As such they often require reconciliation to validate and can "drift" from being accurate over time.
- Contain only a subset of transactional attributes, and thus cannot answer all information needs transactions can.
- Require some level of processing beyond transaction capture.

Posting Keys: How to Create Balances

Posting all attributes takes too long because:

- Too many attributes, with
- Too many values in each attribute, creating
- Too many balances, and
- Too many transaction with no real summarization

Large organizations cannot process the data in time

These are typically the attributes used for critical postings:

- Few attributes, with
- Few values, produce
- Few balances, and
- Few transactions to post after summarization of them

Thus attributes are often used in the general ledger

Customer	Account	Interest			Legal	Cost			Transacti	Source System			
Contract ID	Open Date	Rate	Domicile	Obligation ID	Entity	Center	GL Account	Tran Code	on Date	ID	Currency	Amount	Description
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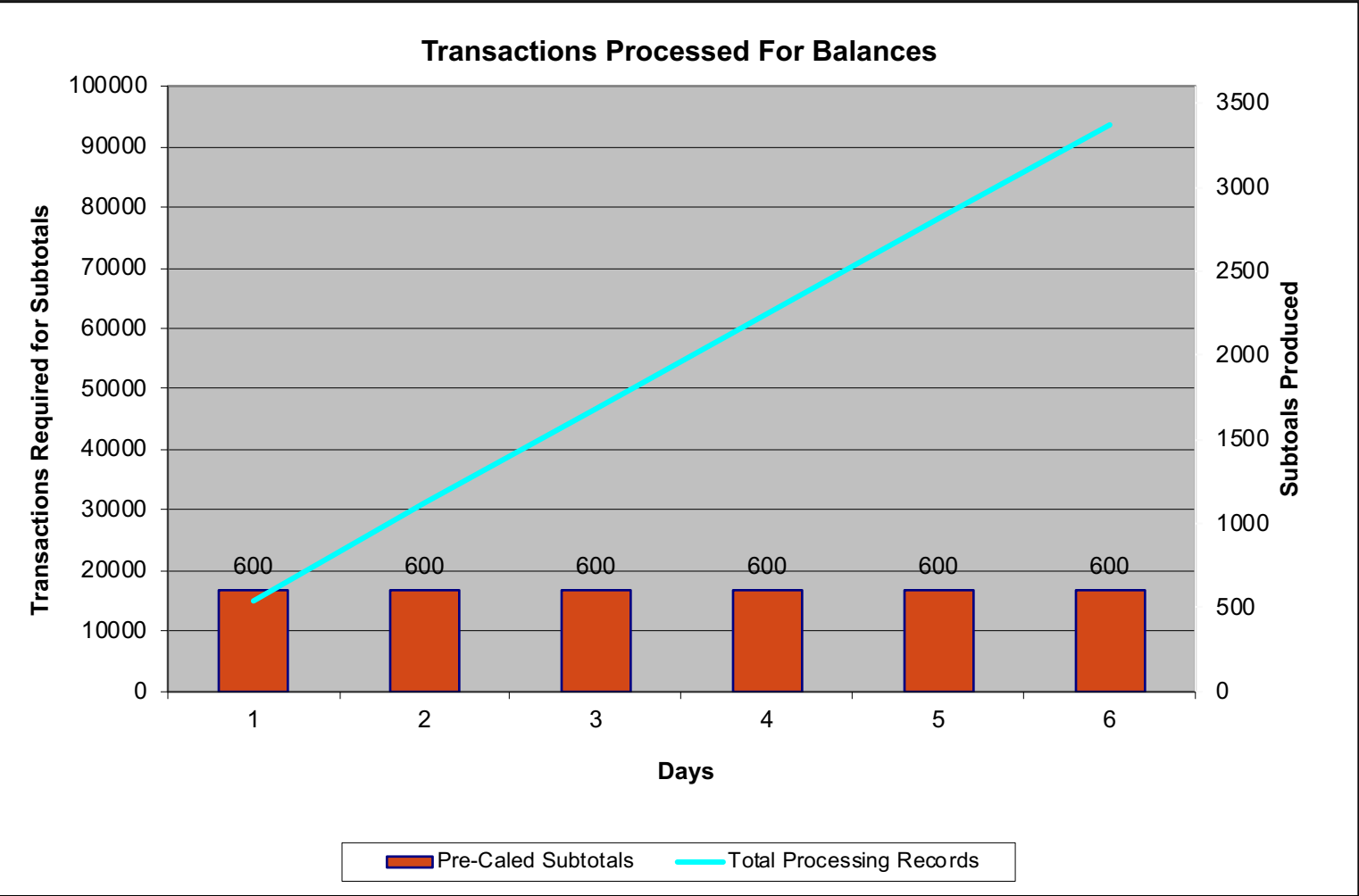
Transactions Required to Produce Subtotals



There are various ways we could produce subtotals from transactions. The most inefficient processing, but lowest reconciliation method way would be:

- Day 1: Use 25 Day 1 transactions 600 times to produce 600 balances. Total: 15,000 transactions (25x600)
- Day 2: Repeat Day 1 with Day 1 transactions, do same thing for Day 2 transactions, then add the resulting balances together. Total: 31,200 (15,000 transactions + 600 balances * 2 days).

The Transaction volumes compound continuously consuming computing resources.



High
Processing
Costs



Low
Reconciliation
Costs

Day	Pre-Caled Subtotals	Trans for Sums	Balances to be Summed	Total Processing Records
1	600	15000		15000
2	600	30000	1200	31200
3	600	45000	1800	46800
4	600	60000	2400	62400
5	600	75000	3000	78000
6	600	90000	3600	93600

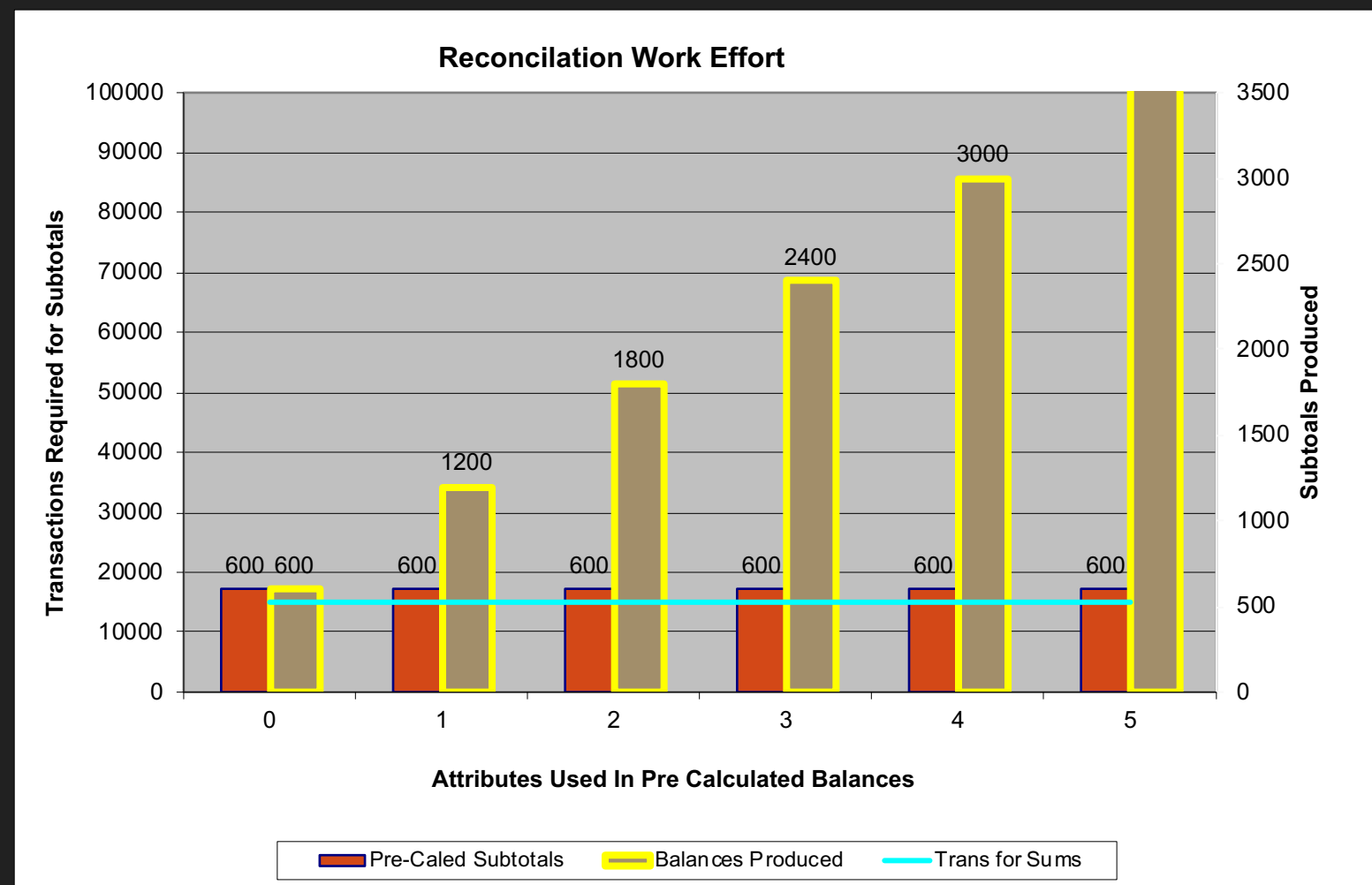
A Posting Process



The most efficient IT method is to:

- Day 1: Use 25 Day 1 transactions 600 times to produce 600 balances. Total: 15,000 transactions (25x600)
- Day 2: Use 25 Day 2 transactions 600 times to make 600 “marginal update” balances, or change in balances from yesterday, and add these to Day 1 balances to make Day 2 balances. Total 15,600 transactions. (25x600 + 600 Day 1 balances)

This is called a posting process, and it uses the lowest amount of computing resources to maintain a running total, but higher reconciliation requirements



Low
Processing
Costs



High
Reconciliation
Costs

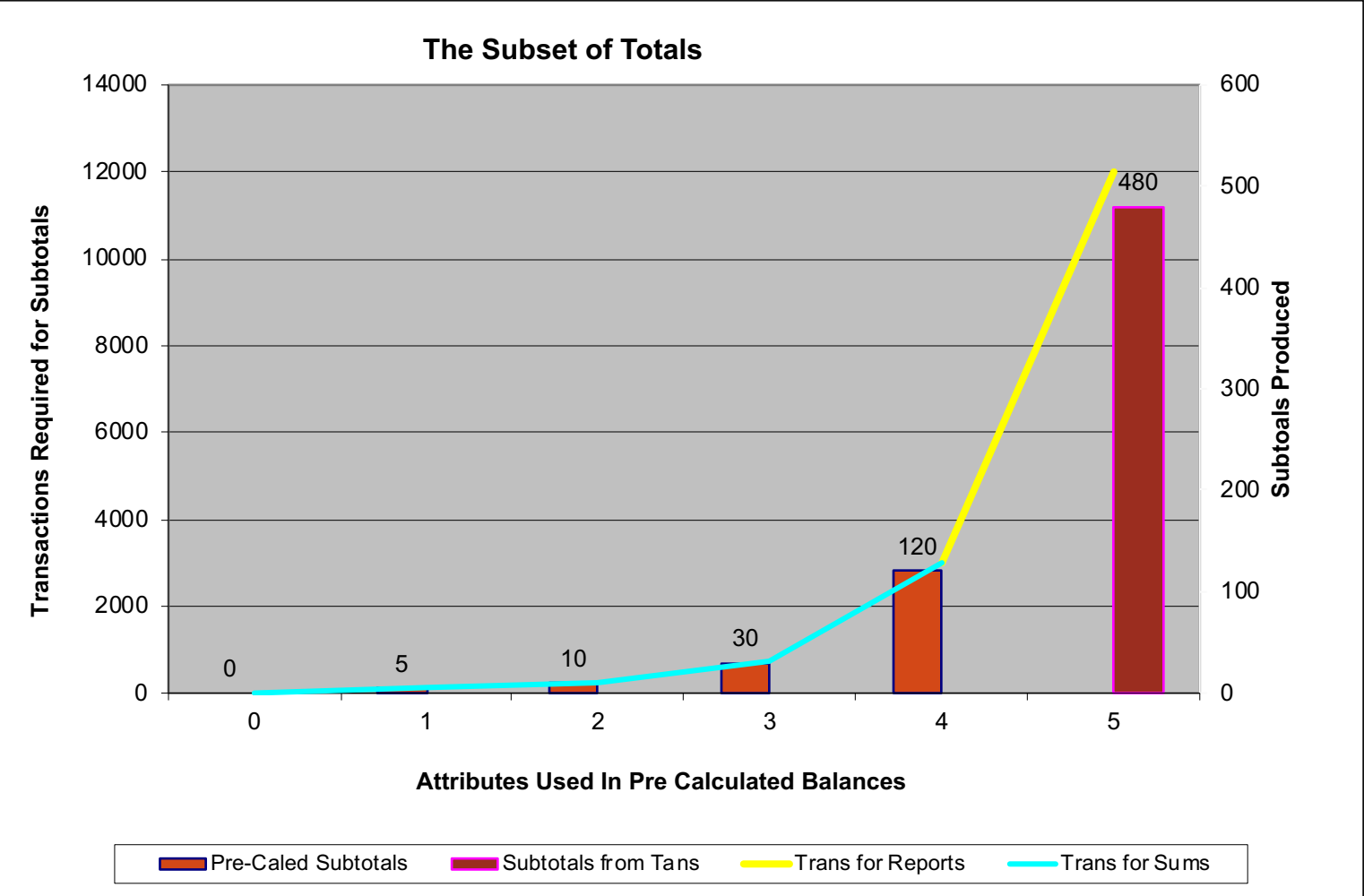
Pre Calced Attributes	Pre-Caled Subtotals	Balances Produced	Trans for Sums
0	600	600	15000
1	600	1200	15000
2	600	1800	15000
3	600	2400	15000
4	600	3000	15000
5	600	3600	15000

A Subset of Subtotals: The General Ledger



Because of constrained computing capacity, no one creates all possible balances. Certain balances may never be used, and thus the compute resources would be wasted.

So, some limited set of subtotals are created very quickly, like the 120 pre-calculated balances, and any other on request balances needed are calculated later. If only 200 of those balances are requested, the total required capacity is less than would be necessary if all balances were pre-calculated.



Pre Calced Attributes	Pre-Caled Subtotals	Subtotals from Tans	Trans for Reports	Trans for Sums
0	0			0
1	5			125
2	10			250
3	30			750
4	120		3000	3000
5		480	12000	

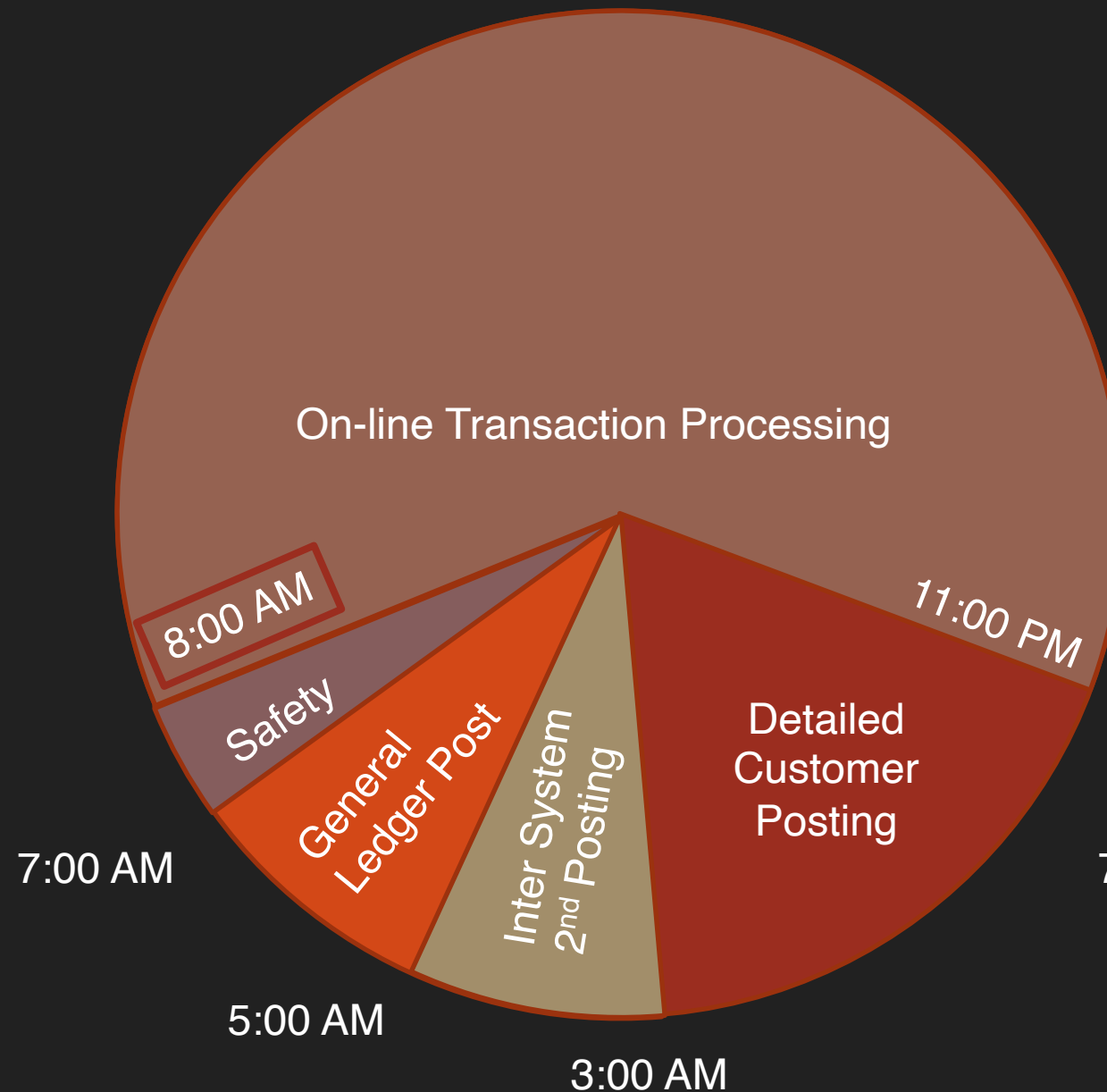
Daily Financial Cycle

The daily financial cycle that drives our aggregation posting processes.

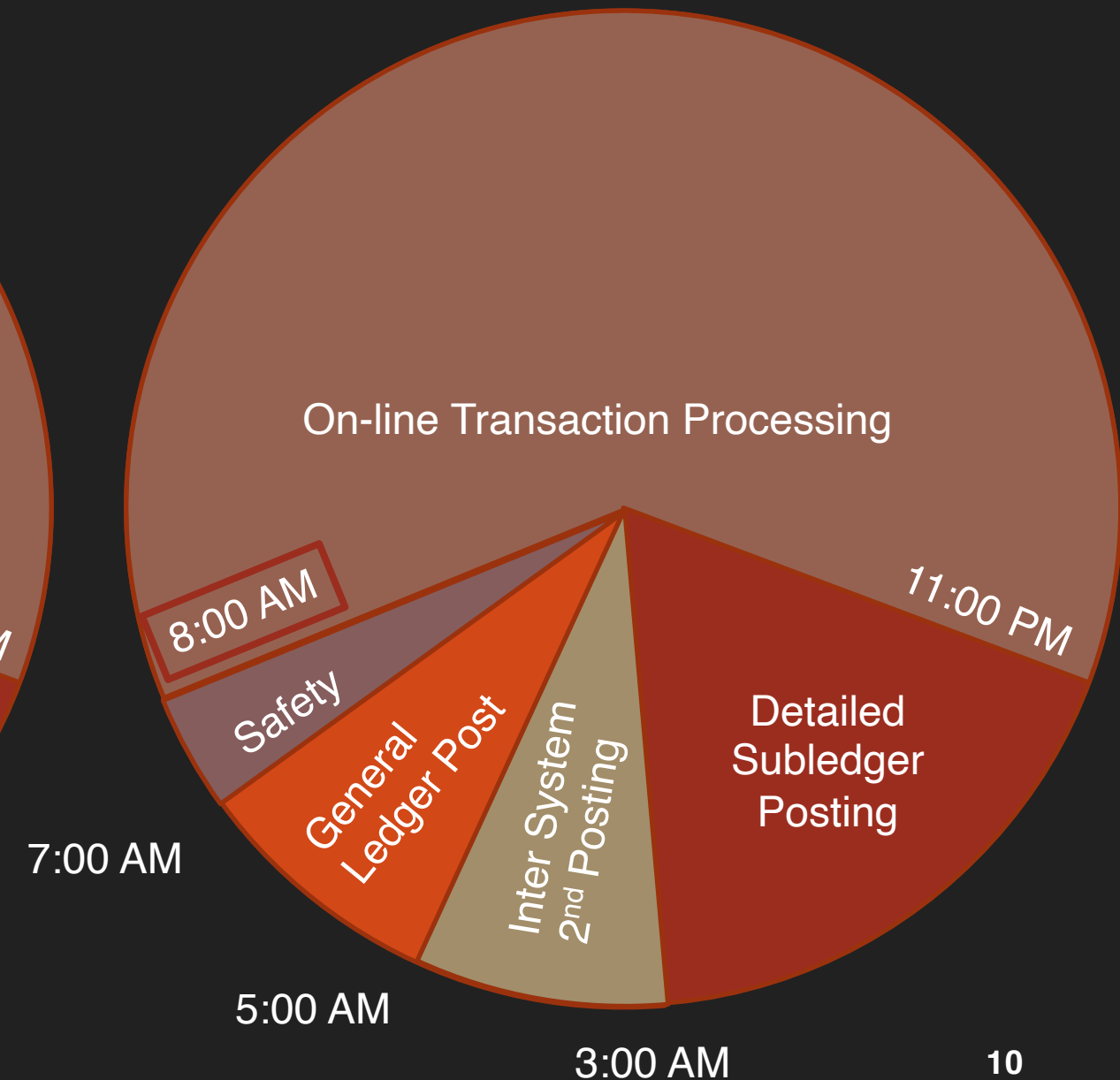
- The originating systems process first
- All partitioned by product, geography, or customers.
- After that, the enterprise financial perspective must be updated, available by next day business opening.

This need drives the General Ledger to be highly aggregated.

A Typical Banking System Day



A Typical ERP System Day



Multiple Data Supply Chains

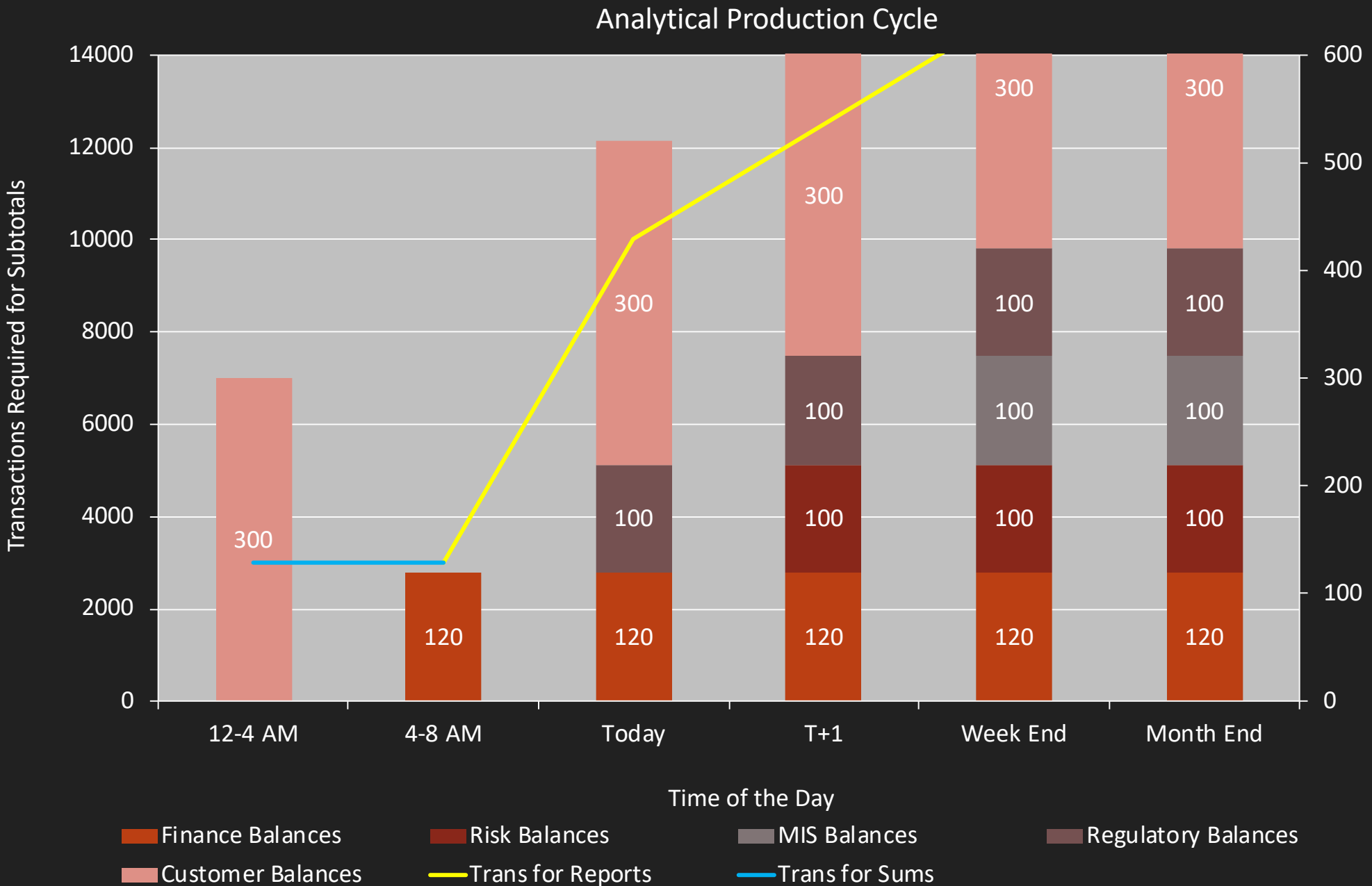


Because of this first, critical fork in the road, other forks were easier to create, for other subject areas, including Risk, MIS, Regulatory, and Customer Profitability processes.

There are many follow-on effects of this of these conscious choices, starting with the impact upon reconciliation, which affect all the balances, being off the chart.

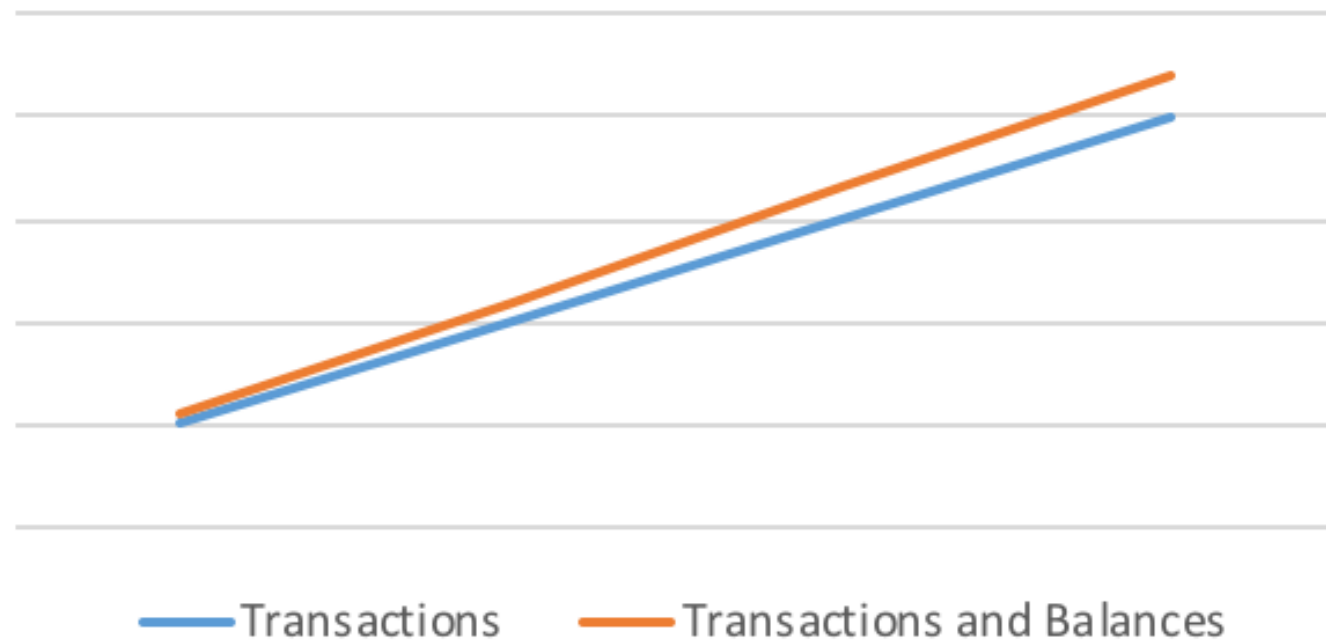
Reconciliation is followed by adjustment to multiple data supply chains, and cross wise actions if errors and differences are not reconciled and adjusted.

This is the environment we live with today.



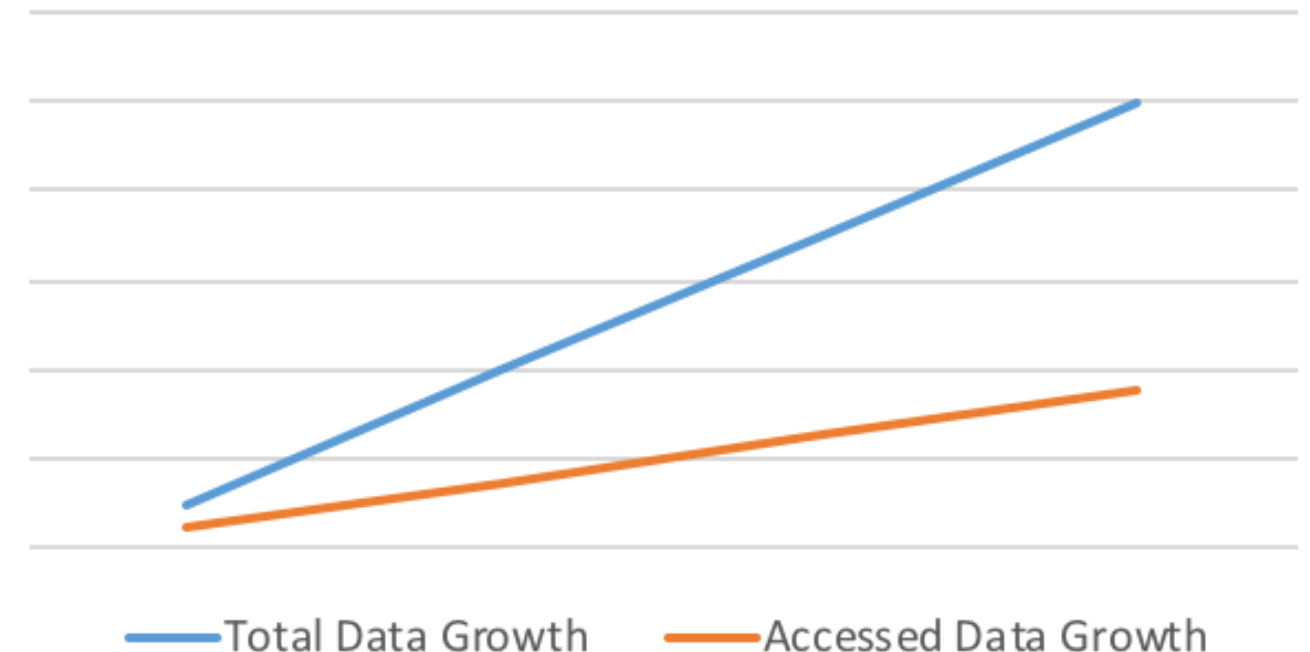
Trends in Data Growth Over Time and Accessed Data verses Total Data

Trends in Data Growth over Time



The use of ledgers, and their associated balances means that in addition to the transaction data, balances have also been maintained. However, because balances are created by selecting a set of transaction attributes as the “posting key,” the volume of balances relative to the originating transactions are much lower.

Accessed Data vs. Total Data

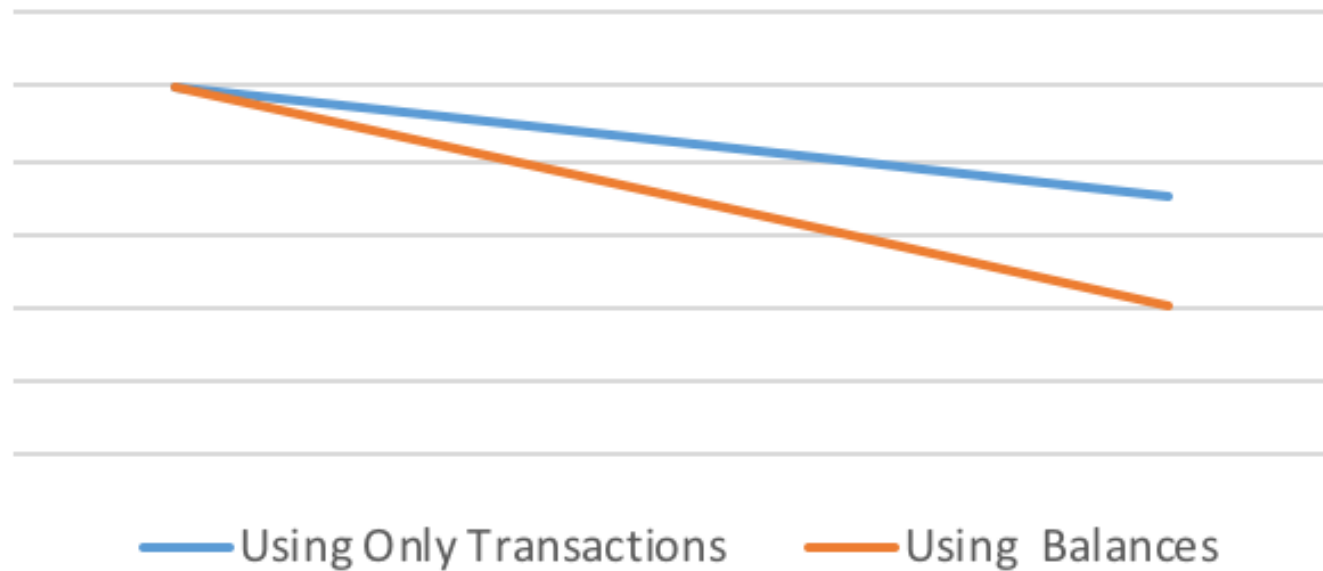


Although slightly adding to costs of total data, the creation of balances significantly reduced the data typically accessed to answer any analytical question; if balances had not been created, much more transactional data would have been needed to continually produce needed positions over time.

Computing Costs Using Transactions and Balances and Balances and Reconciliations



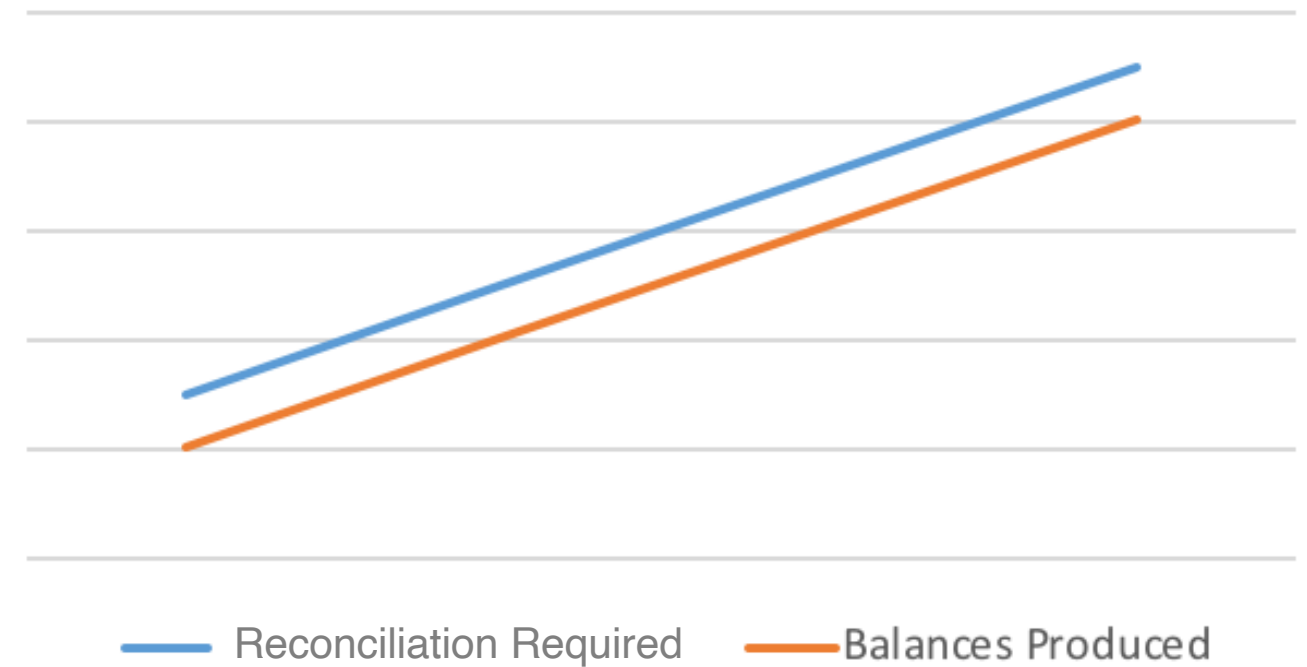
Computing Costs Using Transactions and Balances Over Time



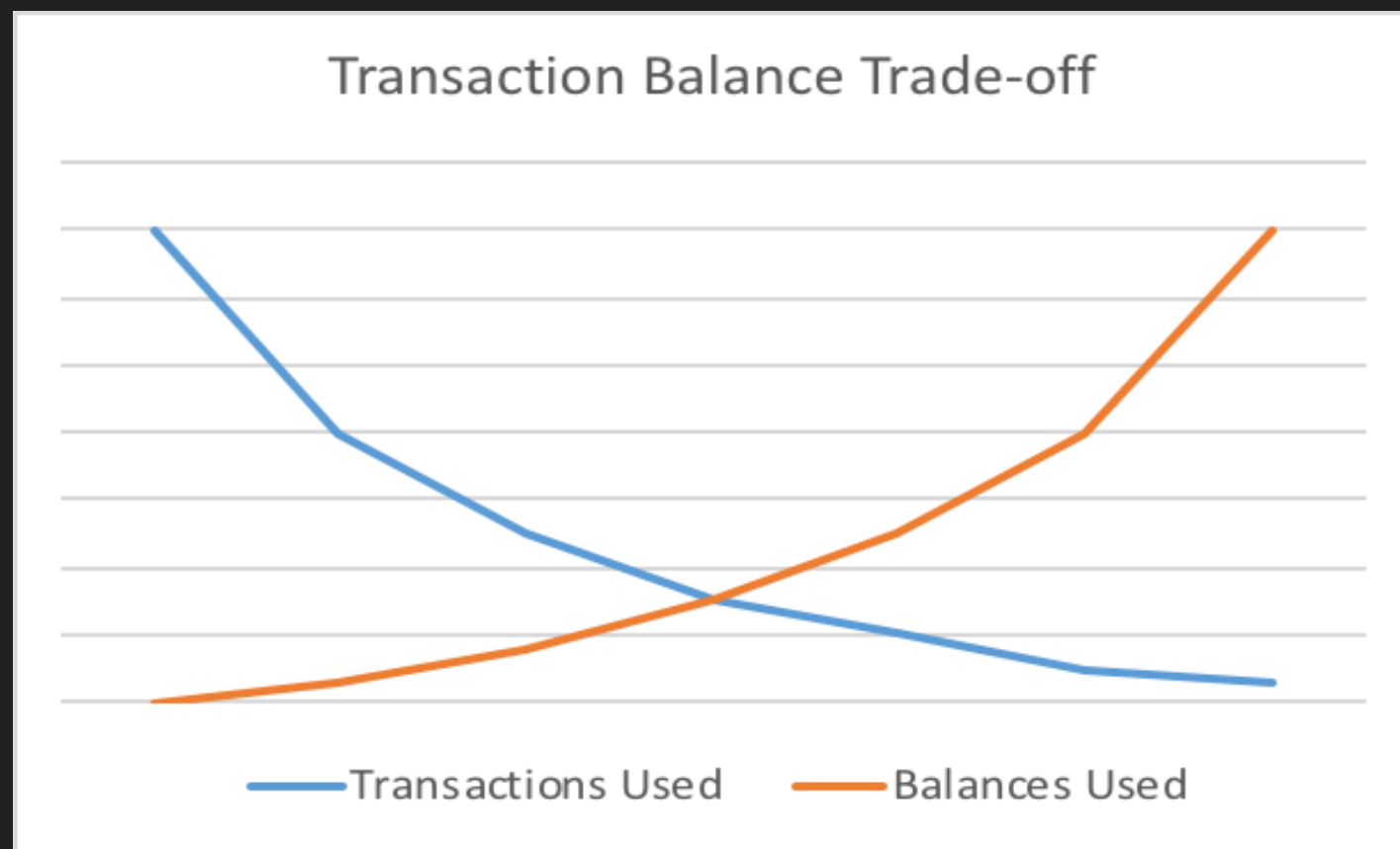
If we hold constant the set of analytical outputs to be produced from a set of transactions, the reduction in computing costs would therefore also be a downward sloping curve.

Consider, though, that if our systems did not produce balances, the increasing number of transactions required over time to produce the same analytical outputs would have flattened the decreasing cost curve.

Balances and Reconciliations



Reconciliation costs increase with each balance produced, to ensure that the balance produced is accurate and remains so over time. Thus, although balances reduce the production of analytical outputs, there is a cost for doing so, and the more balances that are produced, the more reconciliation that is required, meaning the declining cost of balances is offset by the increasing cost of reconciliation

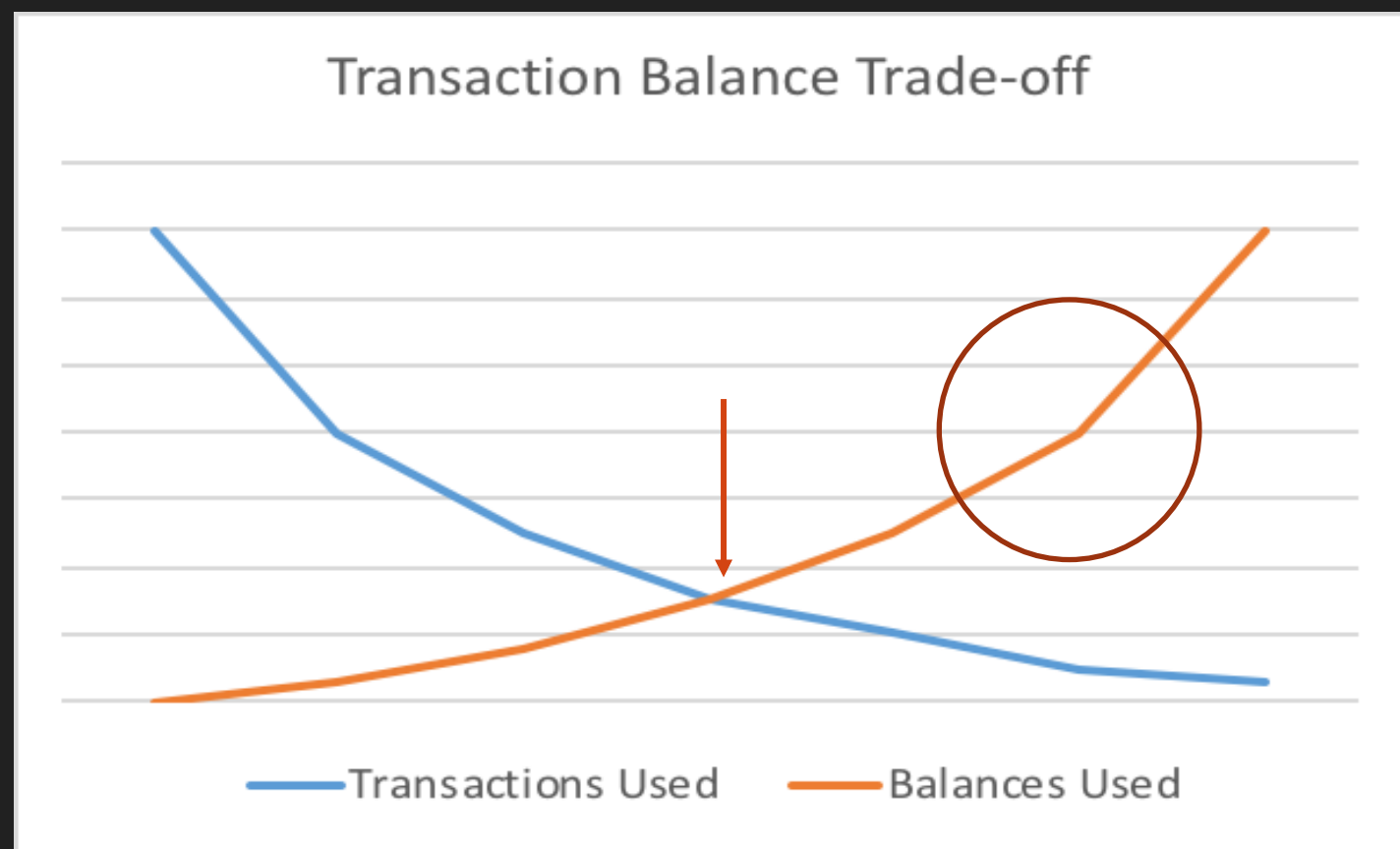


Total cost on the X axis represents cost to:

1. aggregate transactions to present positions,
2. whether done when required for a query or as part of the transaction capture/posting process
3. and cost to present the position to the user.

The Y axis is an isoquant analysis of the costs to produce the reporting result set using the combined data; on the Y axis far left, the set of queries are produced only using transaction data. The next position on the graph slightly to the right is using mostly transaction data, but also utilizing a small number of balances.

The point where balances cross transactions points to the minimum costs of maintaining financial data for the given set of reports or queries. This point uses a set of balances to answer those queries which they can, but also uses transactions to answer those which balances to not satisfy.



The far right of the graph is a possible system condition because balances can be created which do not reduce the total cost of answering questions. This can occur because:

(1) The number of balances (permutations) that can be produced from a set of transactions can exceed the number of transactions involved.

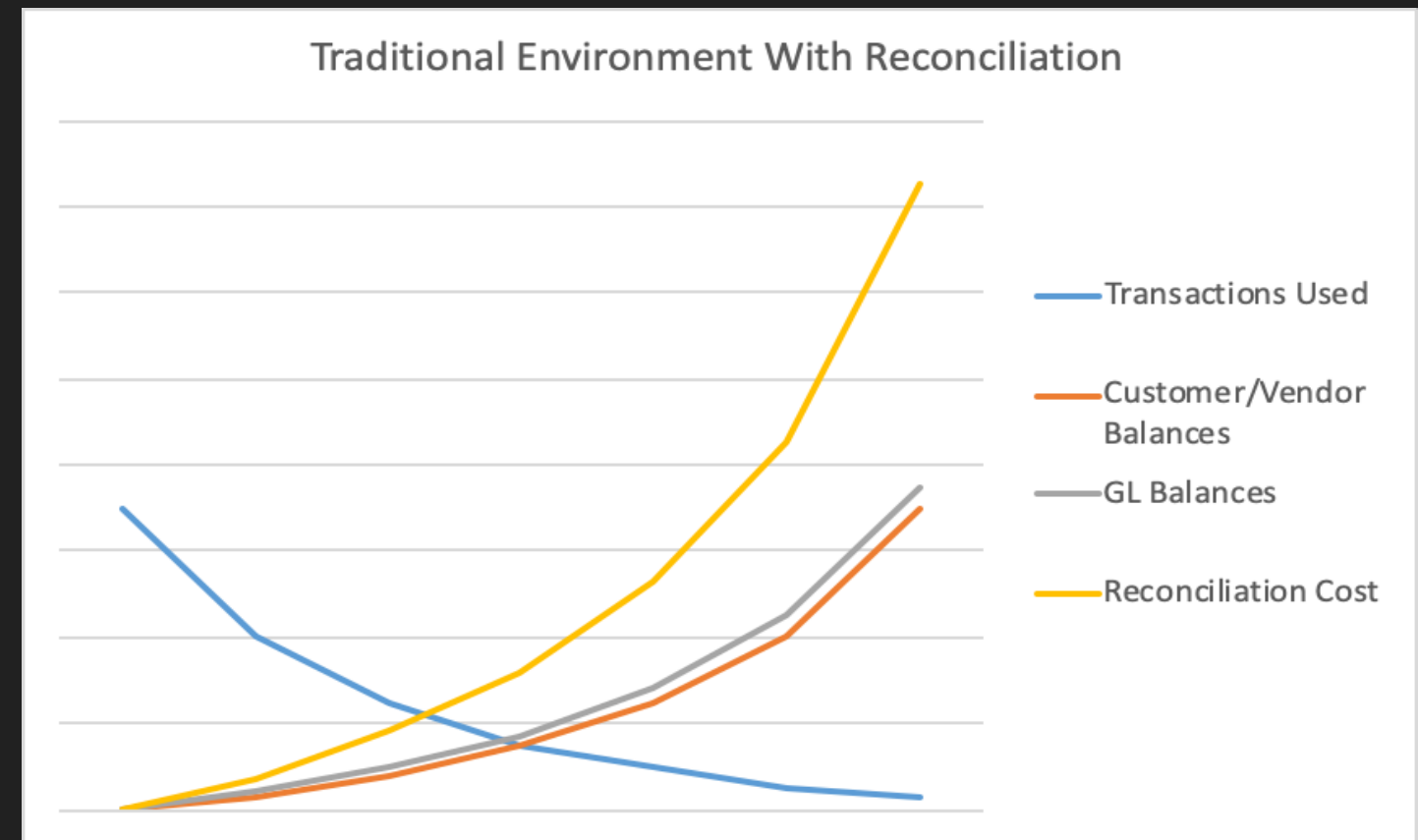
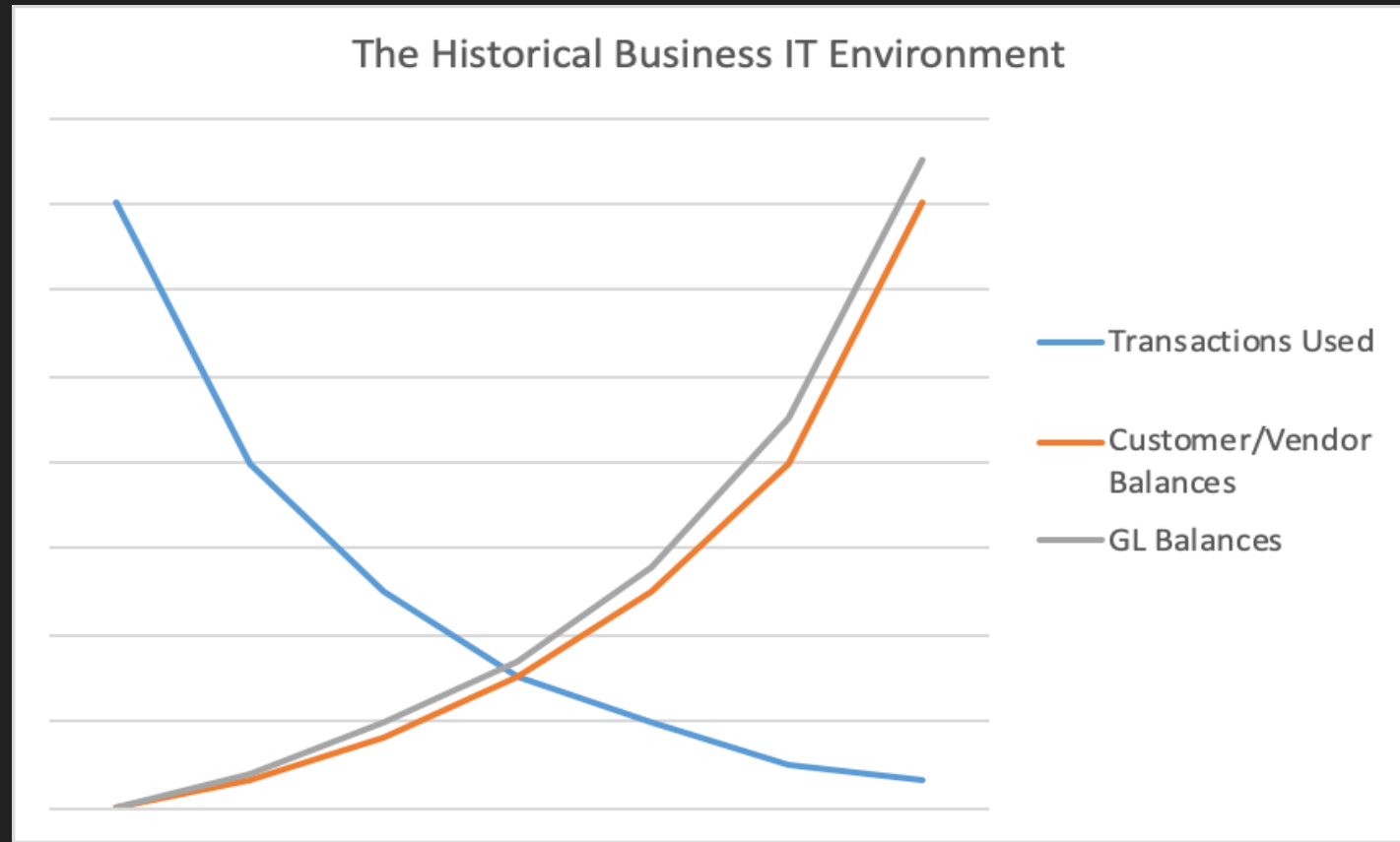
It is a factorial of number of possible attribute values in each field determines the possible balance permutations.

(2) Balances can be produced by systems which are never requested by end users. If a question is asked only once, it would be less expensive to create these needed positions from transactions when requested than incurring the fixed cost of producing these balances systematically in the posting process.

(3) The increasing reconciliation costs accelerates the costs of balance production.

Because compute costs for analytical processes are highly dependent upon volumes of data, it is likely the lowest cost point also happens to be the point of the lowest volume of accessed data needed to answer the given set of queries, with the fewest number of balances which will require the lowest total reconciliations.

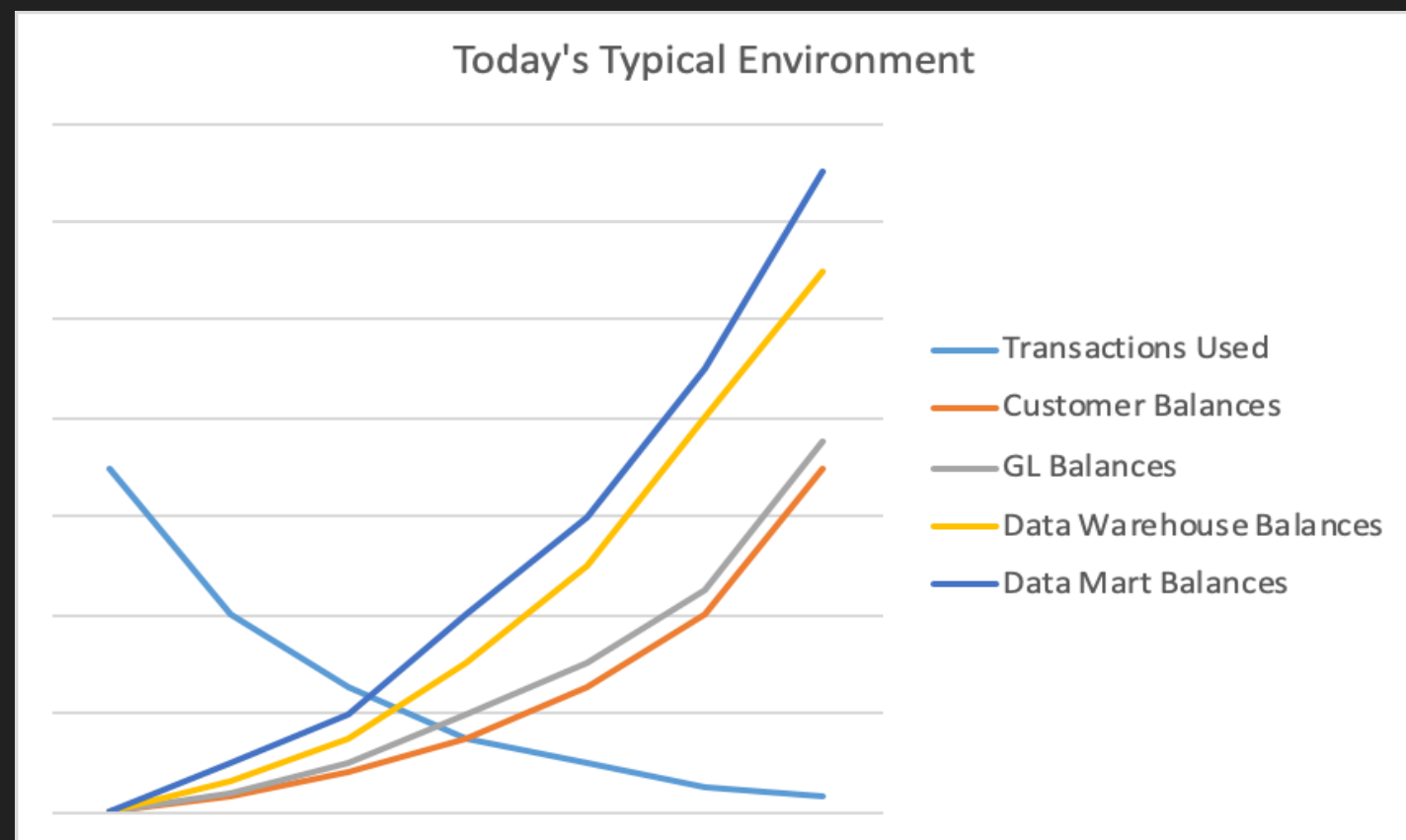
GL Balances are Inadequate



The desired point is not likely arrived at by keeping the same set of balances developed through the decades of system development, namely:

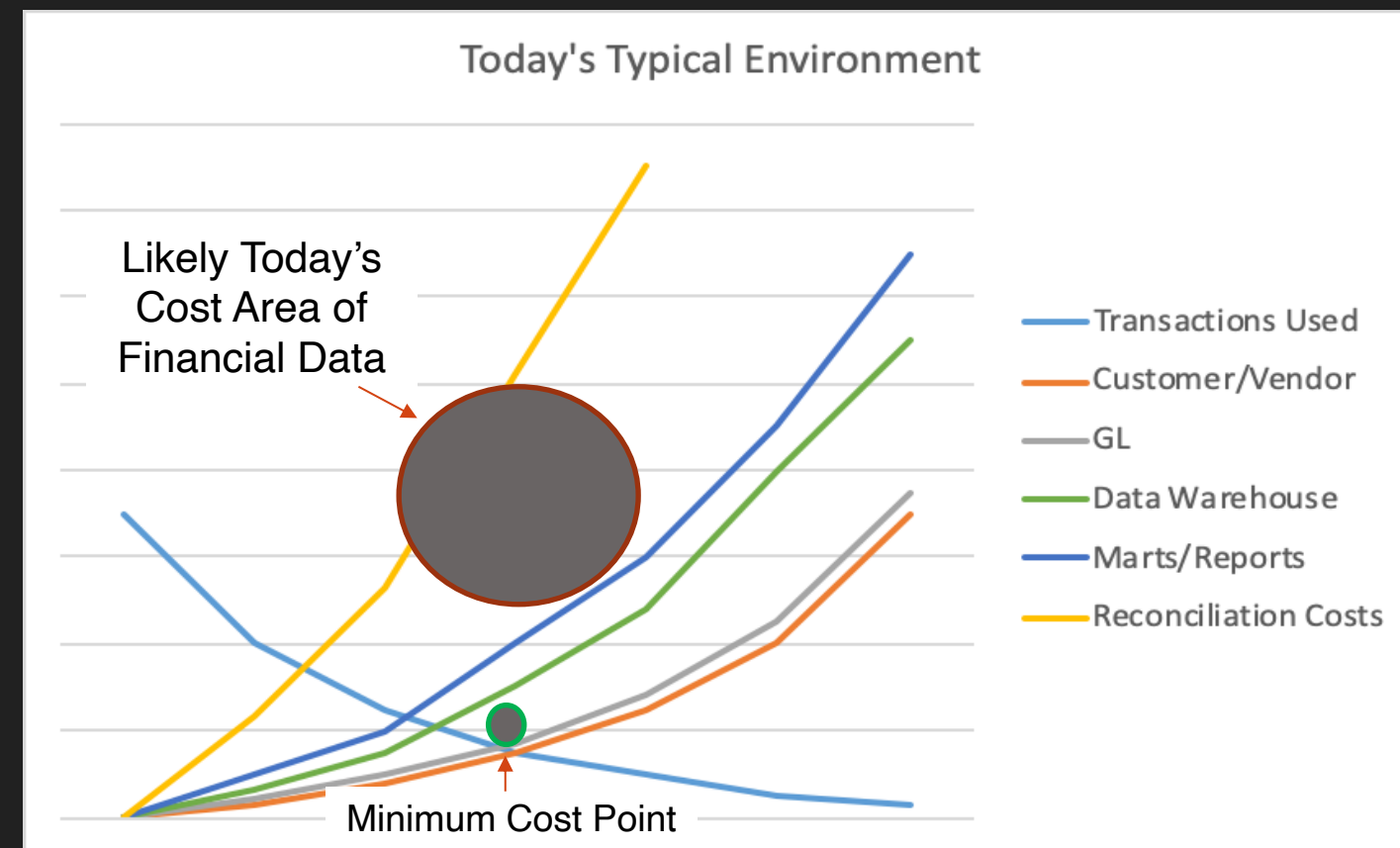
- (1) Customer / vendor / employee account balances kept in the product or source ledgers
- (2) Enterprise balances, usually termed the General Ledger balances.

In the typical periodic (daily in most cases) financial cycle, the transactions are first posted to the more granular source systems, and then transformed into a highly aggregated but very limited set of enterprise attributes, the General Ledger posting key or enterprise code block.



The GL balances are duplicative of the Customer/Vendor balances in many ways; yet reconciliation of the low-level balances to these balances consumes an inordinate amount of effort to comply with regulatory, audit and financial control processes.

Today's organizations retain many more "balances" in the form of reporting structures than the two from decades ago.



The lower level balances are useful for greater purposes—they approach transactional record reporting flexibility. Broad based usage of the maintained balances is a likely criterion of the balances to be maintained. Through the maintenance of customer / vendor and employee keys on these balances, data normalization rules are more readily applied, thus increasing the attributes available for answering reports from these balances.

IBM Scalable Architecture for Financial Reporting (SAFR) and Svayam Infoware

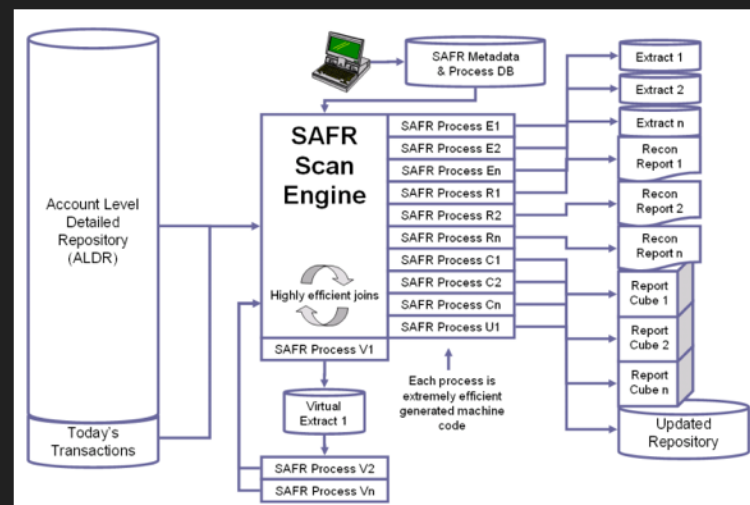
Financial Period End Management Solution

An Evolving Example of a Universal Ledger

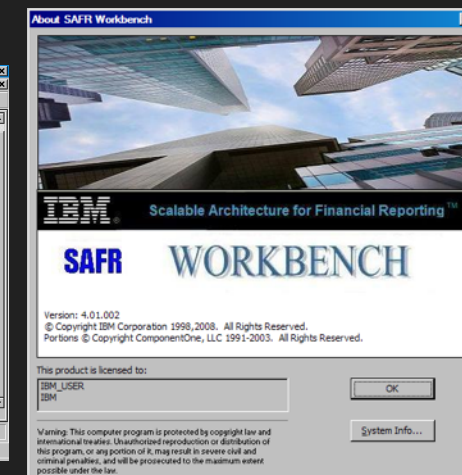
IBM's Scalable Architecture for Financial Reporting™ SAFR



- What is SAFR
 - Software from IBM
 - Generates highly efficient machine code for tremendous scalability
 - Can be an ETL (Extract Transform and Load) tool but for “reporting,” and extract processes
 - Or can be an application develop engine
 - Created for reporting processes for world's the largest financial organization
- What it is Not
 - A database or database centric technology
 - An end-user query tool
- What it Does
 - Rapidly produce many outputs in one “pass” through a database
- Why is it Important
 - Allows data analysis at very low levels of detail
 - And of many more attributes and various coding structures including legacy and transitional code blocks
 - With greater history
 - To support many more financial, regulatory, management and even risk analysis outputs
 - Which all reconcile, are consistent and have the same cut off
- What are the Business Benefits
 - Reconcile source data prior to discovering issues in production
 - Great transparency
 - No charge for development system use with consulting systems engagement



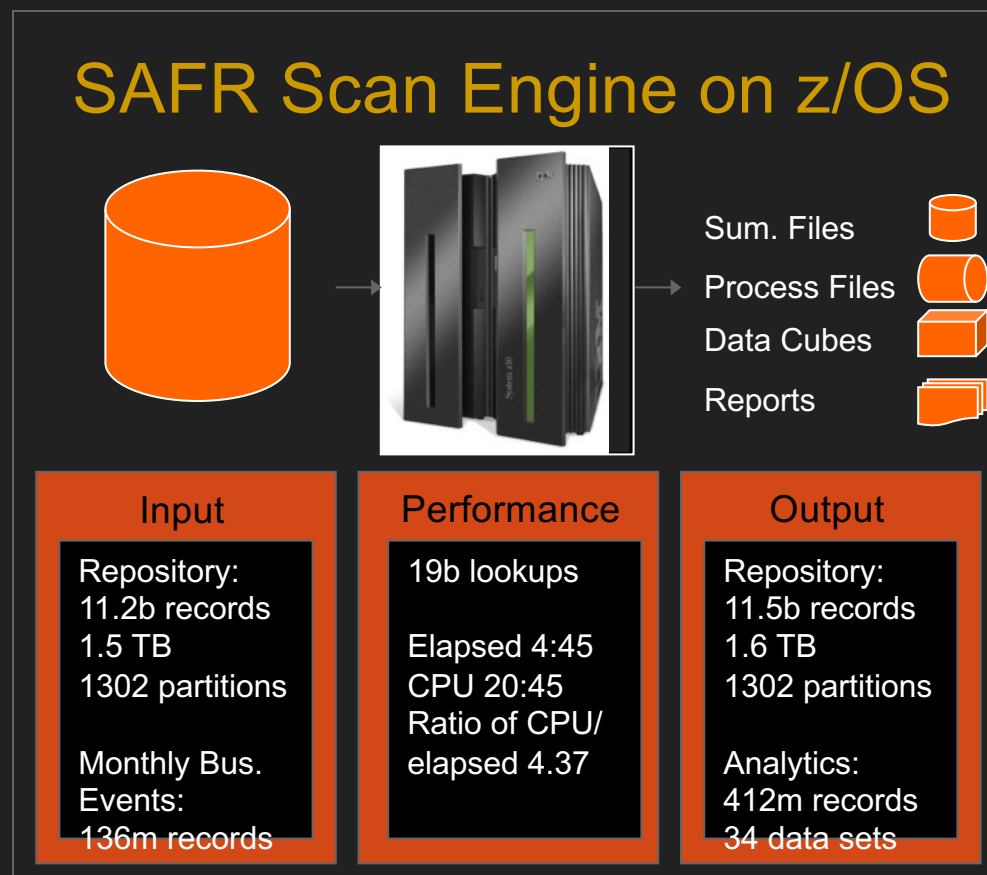
Column Source	Column Source Type	Column Source Value	Effective Date Value	Lookup Path	Lookup Field	Data Type	Length	Decimal Places	Scaling Factor	Sign	Numeric Mask	Format-Phase Calculation	Record Aggregation Function	Group Aggregation Function	Legend
PRODUCT_CODE	PRODUCT_CODE	PRODUCT_CODE				Alphanumeric	10	0	0						PRODUCT_CODE
PRODUCT_DESCRIPTION	PRODUCT_DESCRIPTION	PRODUCT_DESCRIPTION				Alphanumeric	50	0	0						PRODUCT_DESCRIPTION
PRODUCT_TYPE_CODE	PRODUCT_TYPE_CODE	PRODUCT_TYPE_CODE				Alphanumeric	10	0	0						PRODUCT_TYPE_CODE
PRODUCT_TYPE_GENDES	PRODUCT_TYPE_GENDES	PRODUCT_TYPE_GENDES				Alphanumeric	10	0	0						PRODUCT_TYPE_GENDES
PRICE	PRICE	PRICE				Packed	10	2	0						PRICE



SAFR is a high performance scan, join and aggregation engine centered on z and z/OS allowing the largest organizations to keep, consolidate and create analytics from business events.

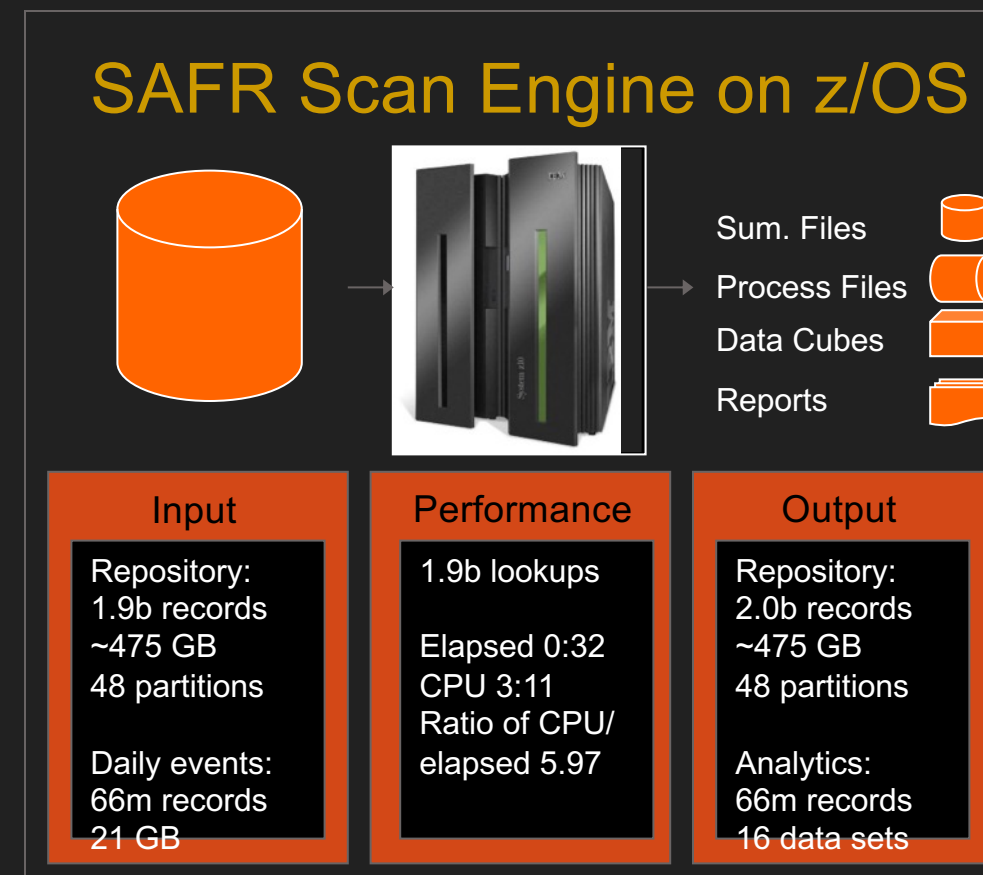
Large US Insurance

Enterprise wide policy, loss, and financial system repositories containing 13 years of business event level detail



Global Bank

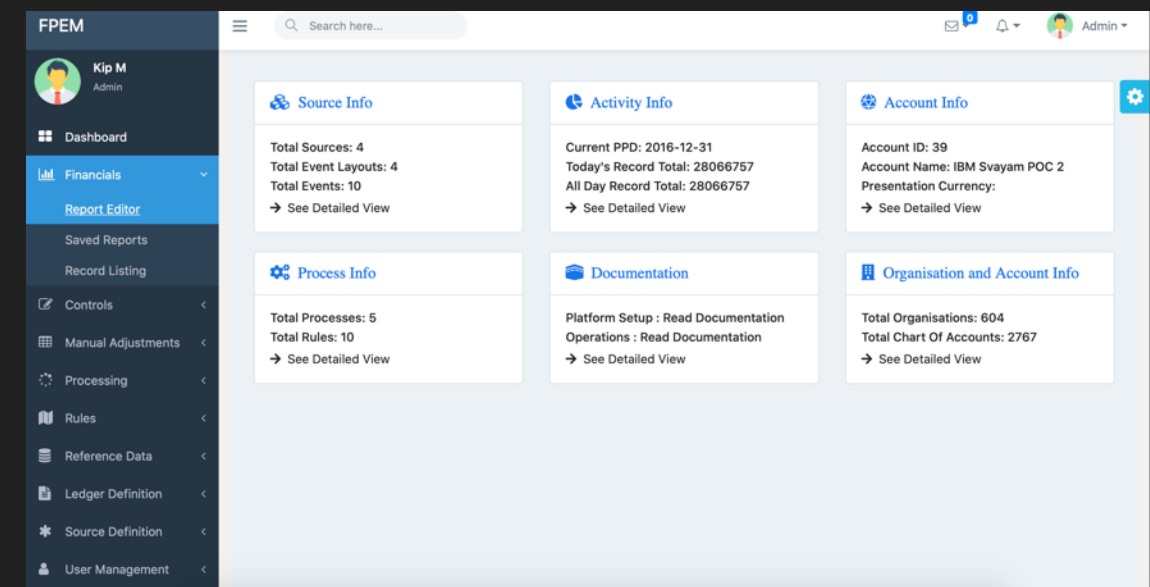
Deploying enterprise wide “data backbone” of controlled, reconciled financial data ultimately supporting customer / contract level analytics



Svayam Infoware: Start-up Focused On This Space

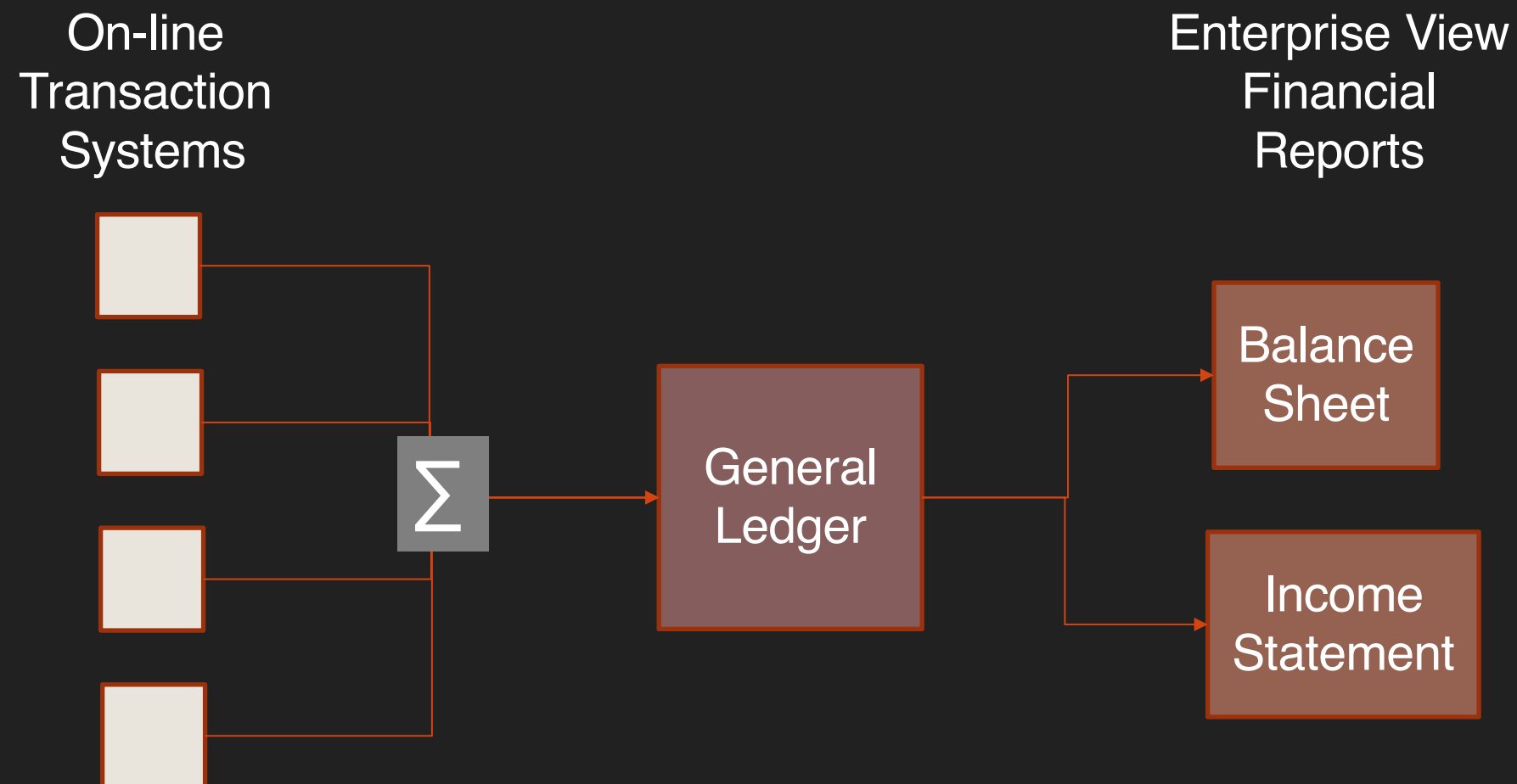


- **Cloud** – Deployable to private, hybrid, or public cloud environments
- **Big Data** – Scale achieved through potential massive parallelization when required
- **Digital** – Fully automated, no manual processes
- **Metrics Engine** – Calculation of myriad of critical business metrics through flexible reporting
- **Instrument Ledger** – Integrated General and Subledger Functionality
- **Daily Close** – Period tracking made simple
- **Rule Based Accounting** – Integration with existing ledgers possible to get started
- **Real Time** – Streaming posting processes supported
- **Straight Through Processing** – End-to-end platform from data ingestion to report delivery
- **Event Driven** – Organization using business events enhances enterprise understanding of data

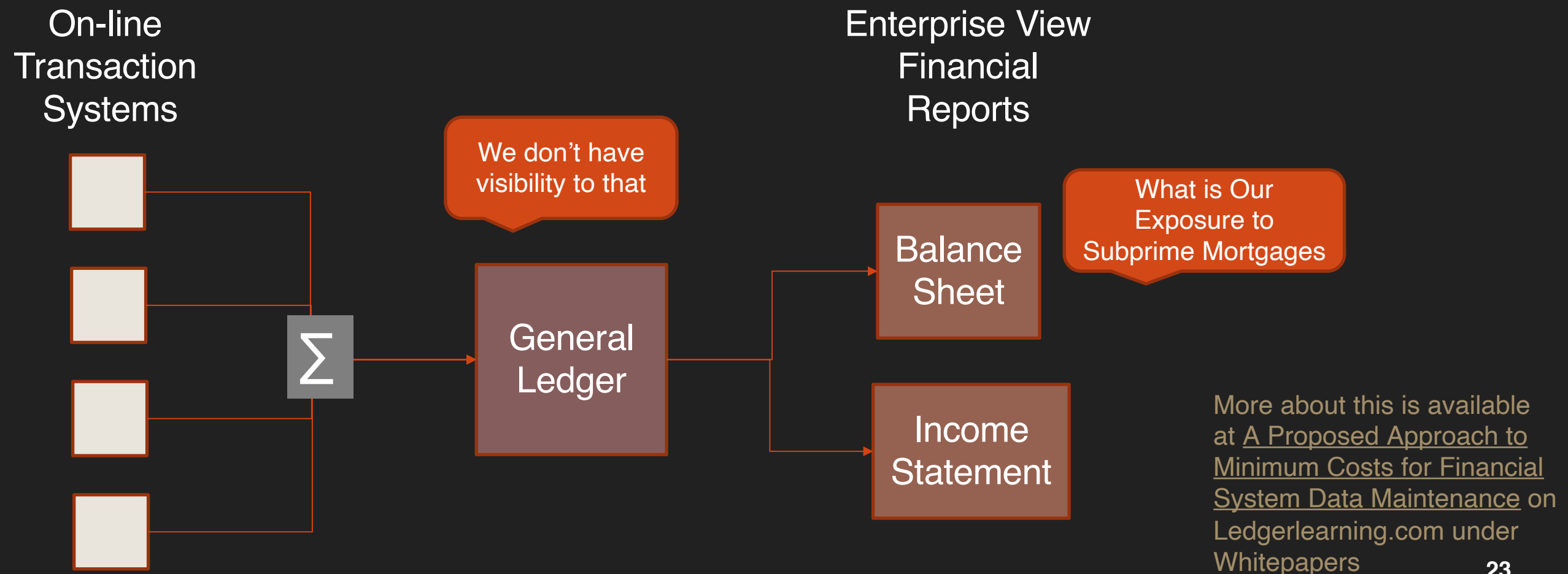


Video Demo of Platform: <https://youtu.be/1jKN5jtkx2U>

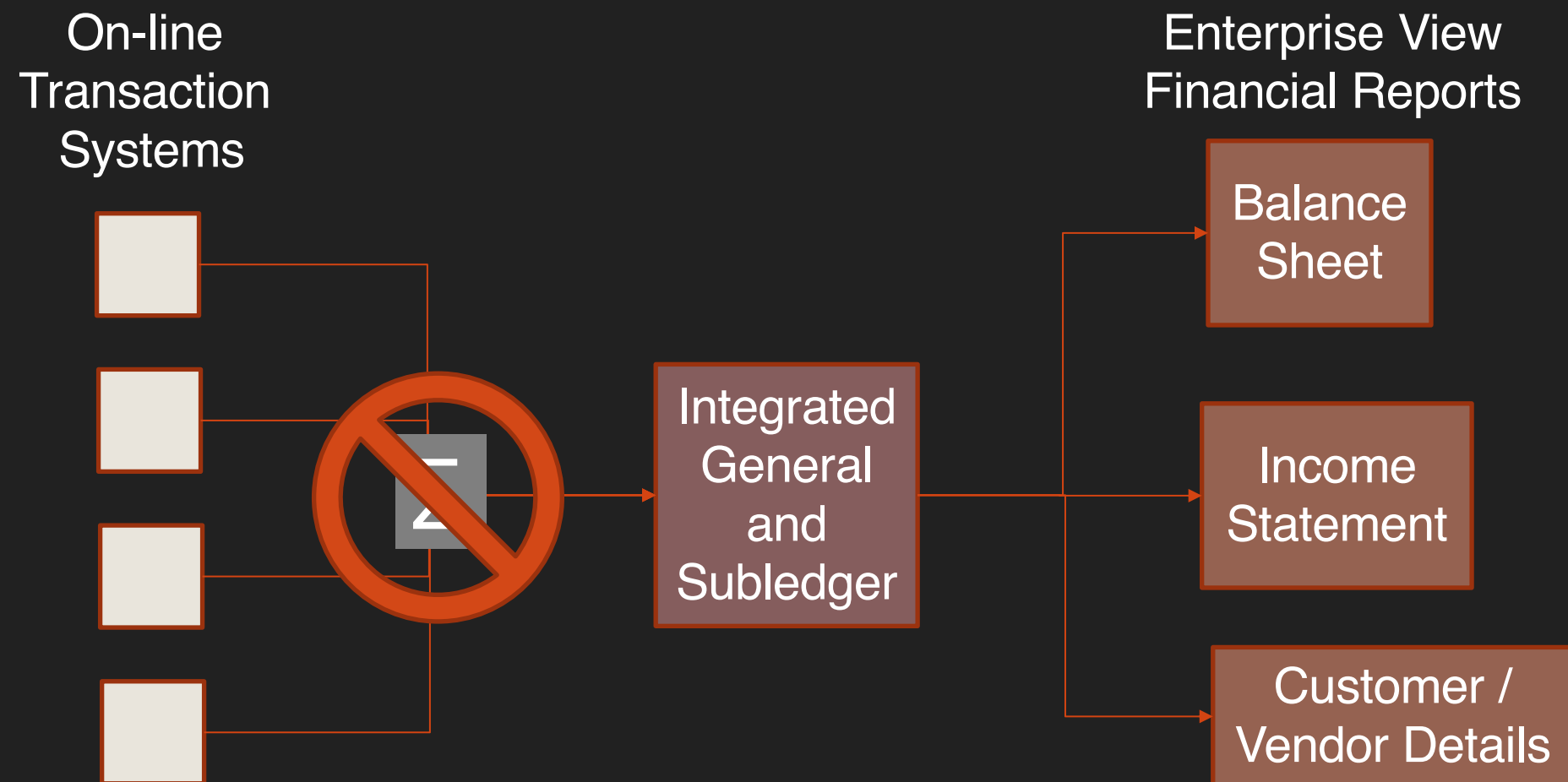
- Every Business must accumulate all the business events, captured in on-line, customer facing transactions systems, into an enterprise view in the balance sheet and income statement
- Historically, this was done by highly aggregating customer/vendor transactions in a General Ledger



- The problem with this approach is that questions asked of the enterprise view cannot easily be answered, as the data is not contained in the General Ledger
- For example in 2008 Lehman could not explain their enterprise exposure to subprime loans
- Costs of not knowing: \$638 Billion Bankruptcy



- With growth in computing capacities, it is now possible to have an integrated General and Subledger environment, called an Instrument Ledger
- The loss of detailed answers through summarization is no longer required
- The Financial Period End Management Platform (FPEM) from Svayam Infoware is the first commercially available package with these characteristics





sharealedger.org

Business Case Development

A Vision Statement March 2020

Mock-ups from Spring 2019

Ongoing Open Data Model Efforts

Recent Personal Financial Ledger Innovations



40 Years Ago, Most People Maintained a Personal Ledger of Some Kind

Bank Ledger



Image courtesy of www.itproportal.com

Monthly Bank Statement

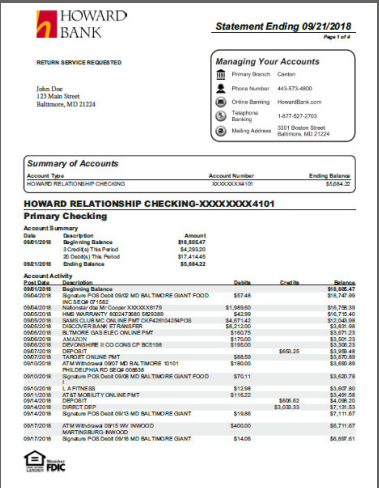


Image courtesy of www.howardbank.com

Personal Ledger



Image courtesy of The Balance.com

20 Years Ago, Many People Started Keeping their Ledgers on the Computer

Bank Ledger



Image courtesy of www.itproportal.com

Automatic Download

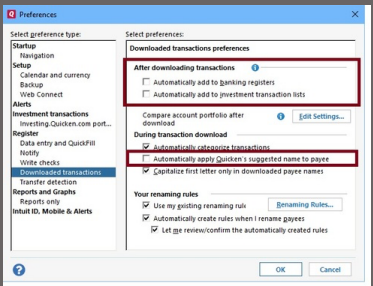


Image courtesy of www.howardbank.com

Starting in the 1990's Quicken Software

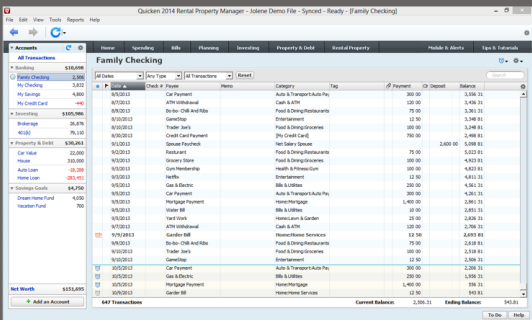


Image courtesy of The Balance.com

Today, Most People Under 40 Do Not Keep a Personal Ledger



Bank Accounts



Investment Accounts



Ledger Mash-up



Credit Cards



Mortgage Account



Consumers Rely Upon Trusted Ledgers From Businesses: Not Blockchain

Under a Traditional ERP System, these are the bookkeeping transactions required for a typical customer/vendor interaction

Customer ERP	Customer Bank	Counter Party Bank	Vendor ERP
1 –Cut Purchase Order			2 - Record Sales Order
4- Receive Goods			3- Ship Goods
5 –Cut Check	8 - Record Withdrawal	7 -Deposit Check	6- Receive Check
10 – Reconcile: a –Bank Statement b – Vendor Statement			9- Produce Customer Statement

With a truly Shared Ledger, these transactions might be reduced to:

- 1. Sales/Purchase Record
- 2. Goods Shipped/Receipt Record
- 3. Payment Record
- (No Statements or Reconciliations)

Potential of 60% Reduction in Transaction Costs

Business Case In Short:

Kip Twitchell Conversation with Claire Zou Price



- The following is a short conversation outlining the Business Case behind a truly Shared Ledger

What problems are you solving?

Building a Universal Ledger

What purpose does the Universal Ledger serve?

To share transaction recording between counter parties, potentially reducing transaction processing costs by 60%.

The knock on effect will be to reorganize ledger structures, which will radically alter analytical processes, for example enabling machine learning against truly trusted data at detailed levels cutting across business processes.

Good value prop. Would it be a replacement or add on of the current system?

Starts as an add-on, to enable the analytical processes, but that doesn't get to the transaction processing savings. Those come through ERP replacement, which is a large cost. But even that can be broken down and implemented by business process.

I like the approach. You can't toss the frog into the boiling water. POC for whom? Like a bank?

The POC is for potential customers. It might be a bank if they want to host the business relationships. But it might start with a large business who could make discounts with vendors possible through shared savings as they use the system as well. It could also be a start-up or a cloud provider as well.

Agreed

60% savings is hard to ignore

Data Focus

- Existing financial system technologies are based upon computing capacities and capabilities available 30 to 40 years ago.
- This project will move shared ledger concepts forward, building upon blockchain's goals, to create an industry basis for a shared ERP environment, at cost-effective scale, starting with general ledger at the core
- This project intends to increase financial system efficiencies, which is a difficult goal given the tremendous focus on cost for years in the existing systems.

Interoperability

- Efficiencies will primarily come through sharing data and processes, which will require standards.
- Making practical standards requires use of them. This project will provide a test environment for developing these standards with multiple organization participation.
- It may result in an initial free service used by individuals and small businesses for sharing financial data and processes.
- And provide a roadmap for growth to larger businesses and for vendors to develop solutions on top of the platform.

Accessibility

- Digitization of currencies leaves large groups of marginalized individuals further isolated from opportunity.
- The ubiquity of computing capacity in phones could be paired with simple financial tools like cash flows, income statements and balance sheets to enable the smallest merchants in the world to join the digital economy and effectively manage their finances.
- Such a platform requires the scale and scope of today's major data platforms, and the accuracy and security of today's financial systems at even lower costs.
- This ambitious scope will require a flexible, multi-party, shared approach.

Shared Ledger

The Ultimate in Efficient Business Systems

First name

Last name

Email

.

Password (6 or more characters)

By clicking Join now, you agree to the LinkedIn User Agreement, Privacy Policy, and Cookie Policy.

Join now

User Name: Kip Twitchell
User ID Number: 089405439-8859

Shared Ledger

The Ultimate in Efficient Business Systems

Personal Fields

New Field

- Category
- Date/Time
- Location
- Company
- Transaction Type

Permissible Values

Field: Category

New Value

- Food and Groceries
- Gas and Auto
- Housing and Rent
- Entertainment
-

Personal Fields are only visible to you, unless you share them in a ledger

User Name: Kip Twitchell
User ID Number: 089405439-8859

Shared Ledger

The Ultimate in Efficient Business Systems

My Ledgers

Ledger Details

Ledger ID: LDR-5694t34-56843

Ledger Name:

Ledger Description:

Posting Type:

Balance Periods:

LDR-5694t34-56843

[The following information was obtained from the records of the Department of Health, Education and Welfare, Division of Child Development Services, Bureau of Early Childhood Development, Office of Special Education.]

Real-time | Periodic

Hourly | Daily | Monthly | Qtrly

- Personal Finances
- Side Business
- Bank Account
- Investment Account
-

User Name: Kip Twitchell
User ID Number: 089405439-8859

Shared Ledger

The Ultimate in Efficient Business Systems

Ledger Participants

Search

- Ned Symmonds ☐
- Tom Hanks ☐
- Fred Horwitz ☒
- Jimmy Hoffa ☐
- Will Accetta ☒
- John Fry ☐
- ☐

Share

Ledger Details

Ledger ID: LDR-5694t34-56843

Ledger Name: Universal Vendor-Customer Ledger

Ledger Description:	A ledger for all shared business transactions, from purchase orders, to shipment and receipts, to invoicing and payments
---------------------	--

Posting Type: Real-time

Balance Periods: Daily

User Name: Kip Twitchell
User ID Number: 089405439-8859
Ledger: Universal Vendor-Customer Ledger
Ledger ID: LDR-5694t34-56843

Shared Ledger

The Ultimate in Efficient Business Systems

Possible Fields

- Category ☐
- Date/Time ☐
- Location ☒
- Company ☒
- Transaction Type ☒
- Amount ☒

Share

Shared Fields

New Field

- Location
- Company
- Transaction Type
- Amount

Shared Fields are visible to all parties in the Ledger

User Name: Kip Twitchell
User ID Number: 089405439-8859
Ledger: Universal Vendor-Customer Ledger
Ledger ID: LDR-5694t34-56843

Shared Ledger
The Ultimate in Efficient Business Systems

Shared Fields

Posted Fields

Stored Balance By: Calculated Balance By:

Location
Company
Transaction Type
Amount

☐
☒
☐
☐☒
☐
☒
☐

Stored Balances designates posting processes.
Calculated balances may be stored based upon usage patterns and efficiencies.

User Name: Kip Twitchell
User ID Number: 089405439-8859
Ledger: Universal Vendor-Customer Ledger
Ledger ID: LDR-5694t34-56843

Shared Ledger

The Ultimate in Efficient Business Systems

Rules for Personal Fields

Field: Category

If Company = "Panera"
And Location Not = "US"
Then
 Category = "Business"
Else
 Category = "Eating Out"

Rules for Shared Fields

Field: Location

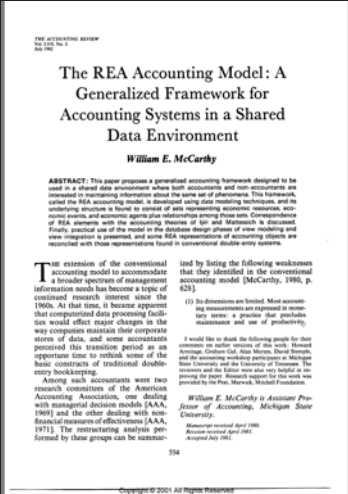
If Transaction Type = "Purchase Order"
Then
 Company = "Mine"

Shared Rules are visible to and agreed by all parties in the Ledger.
Personal Rules are Not.

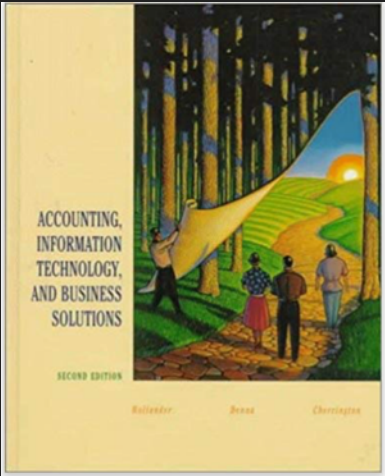
- In March 2017, I was given “Model-Driven Design Using Business Patterns” by Pavel Hruby (2006) by Steve Turner, a Bay Area CPA interested in the REA theory
- That book led to a review of the related materials I have accumulated over the years:



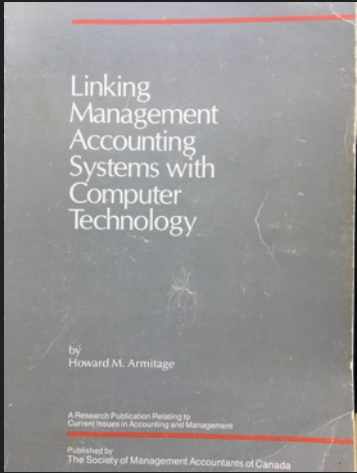
- In February 2020, Dr. McCarthy shared:
- International Organization for Standards (ISO) standard for EDI based upon REA theory
 - A prepublication version of a forthcoming retrospective monograph on REA theory



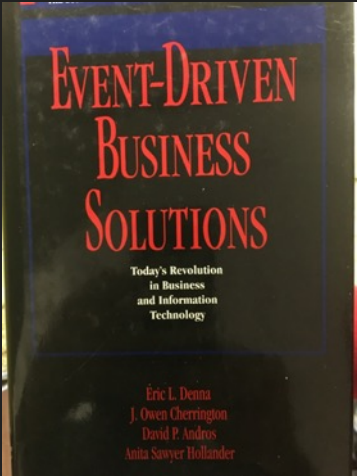
The Original Dr. Bill McCarthy REA paper in Accounting Review, July 1982



Accounting, Information Technology, and Business Solutions 2nd Edition 1999 textbook by Hollander, Denna, Cherrington



“Linking Management Accounting Systems with Computer Technology” 1987 monograph by Howard Armitage



“Event-Driven Business Solutions: Today's Revolution in Business and Information Technology” Eric Denna et. al 1993

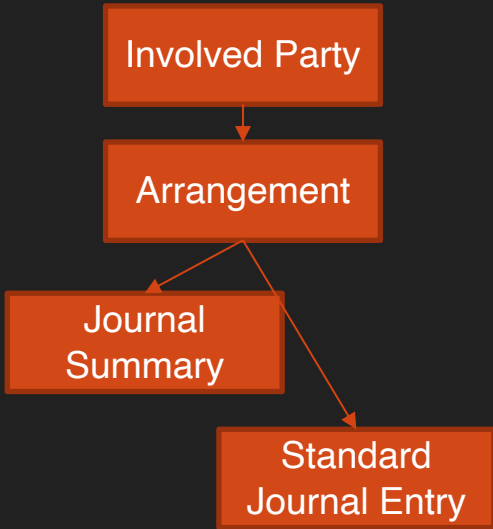
International Organization for Standards ISO/IEC 15944-4:2015(E)
Information technology Business: Operational View part 4: Business transaction scenarios accounting and economic ontology



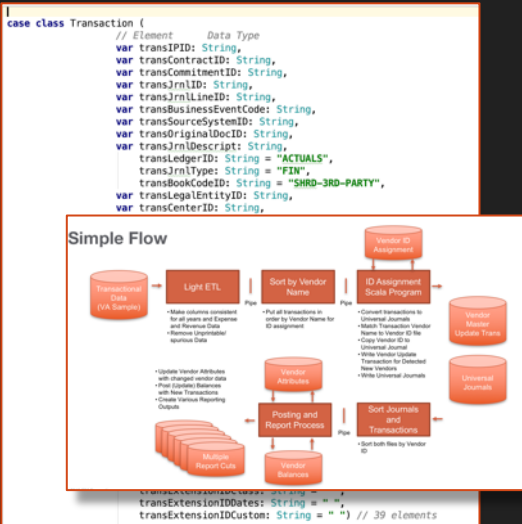
Forthcoming “The REA Accounting Model as an Accounting and Economic Ontology” By William E. McCarthy, Michigan State University, Guido L. Geerts, University of Delaware, Graham Gal, University of Massachusetts



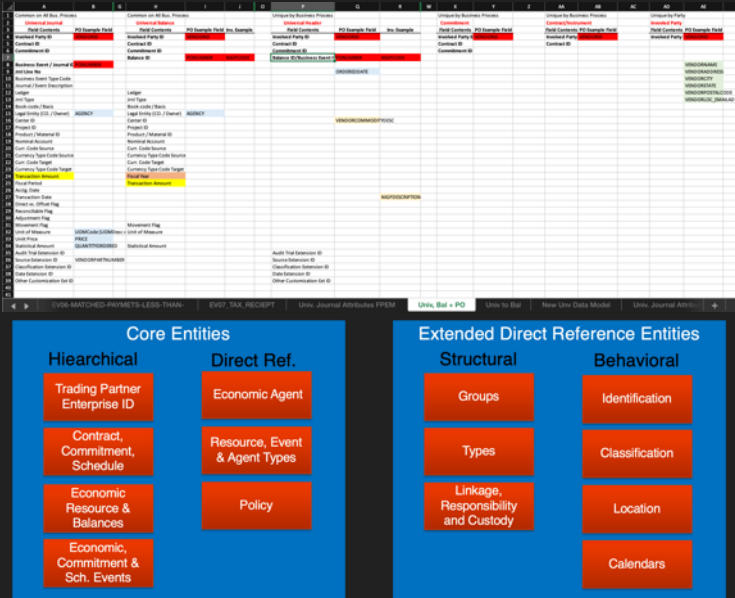
- The following experiences, prototypes and proof of concepts also provide input into the system



FTP System



3rd Quarter
2017 Scala
Posting Demo
System



Taking from Hurby's "Model-Driven Design Using Business Patterns" book (2006), I proposed that the following entities be considered for part of the Universal Ledger. The Hierarchical Entities are for high volume transaction, balance and descriptive records. The others are direct reference, or indexed access, rather than hierarchical.

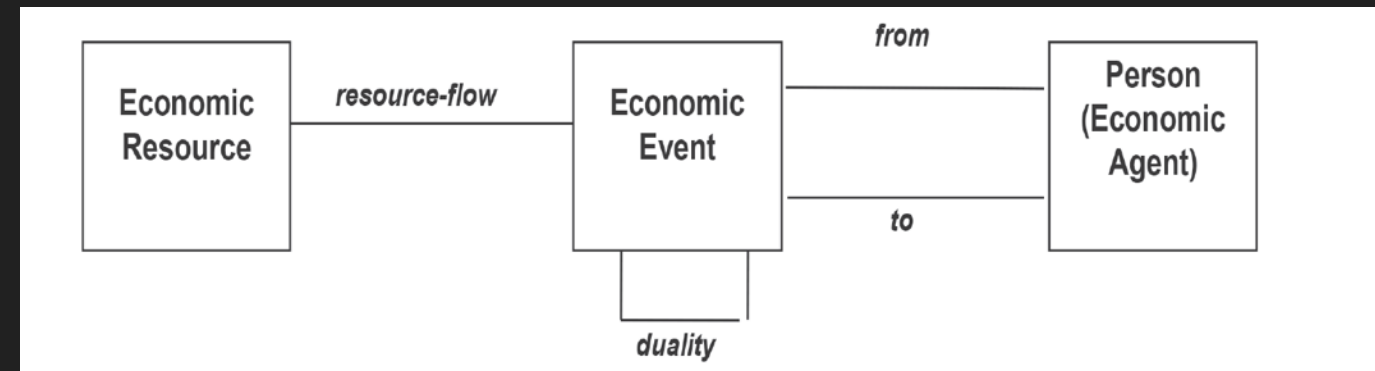
Spring 2018
Academic
Analysis



Fall 2019
Swayam VA
Data POC

Original REA Model

This diagram, which has been used to explain REA, meaning Resources Events and Agents, for perhaps 30 years.



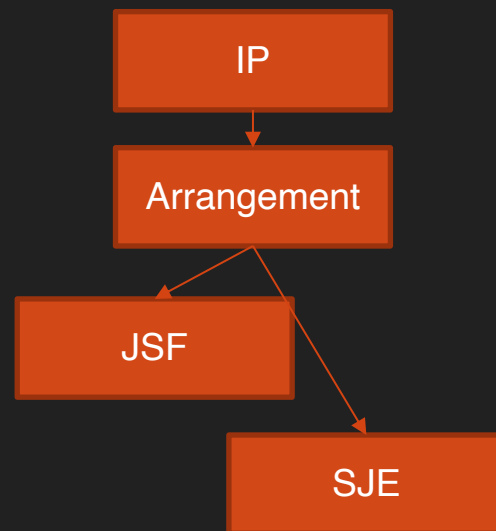
Arrangement

Standard
Journal Entry

Involved
Party

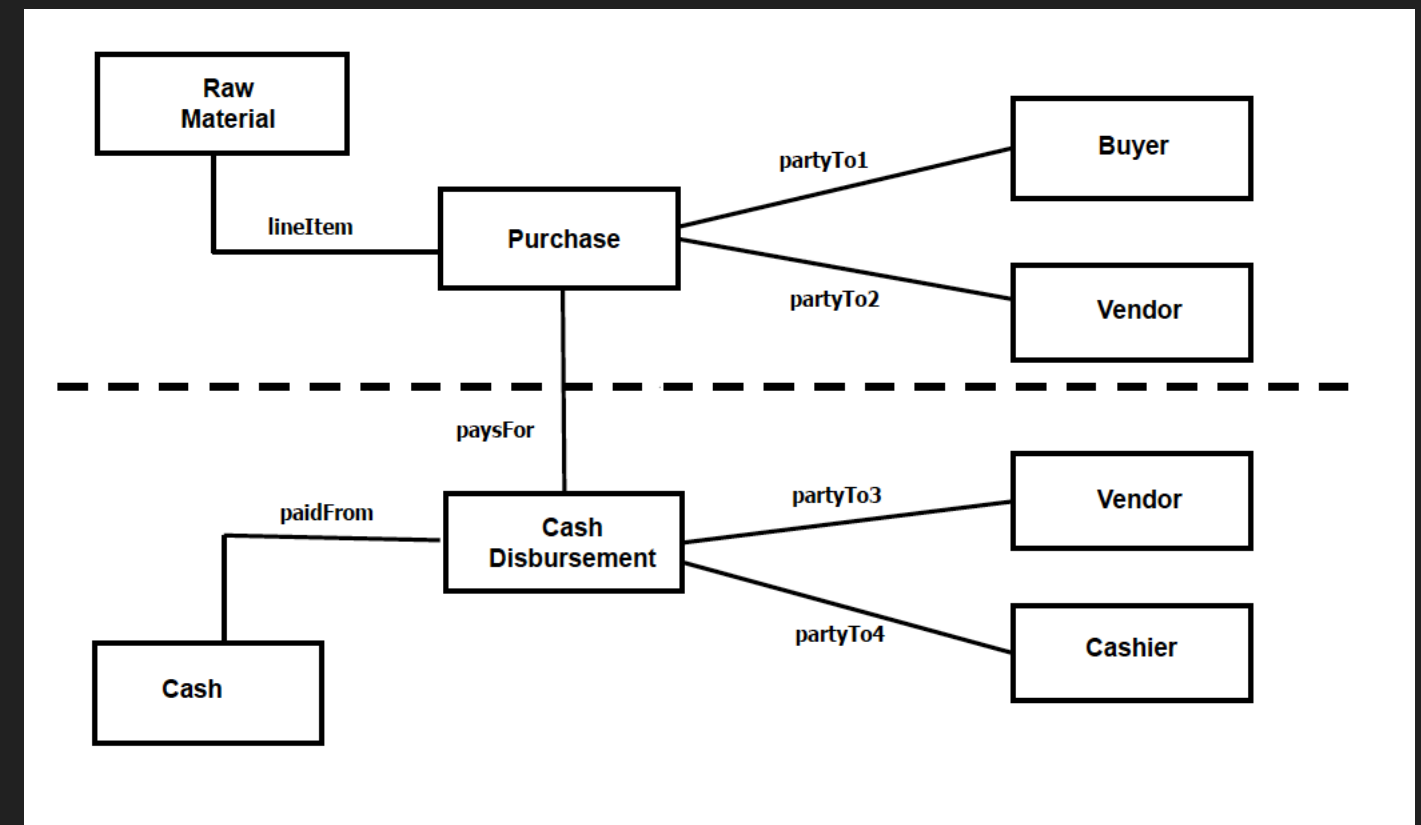
Example
Below

One could claim that the FTP model is quite close to this model, albeit not necessarily intentionally



FTP System

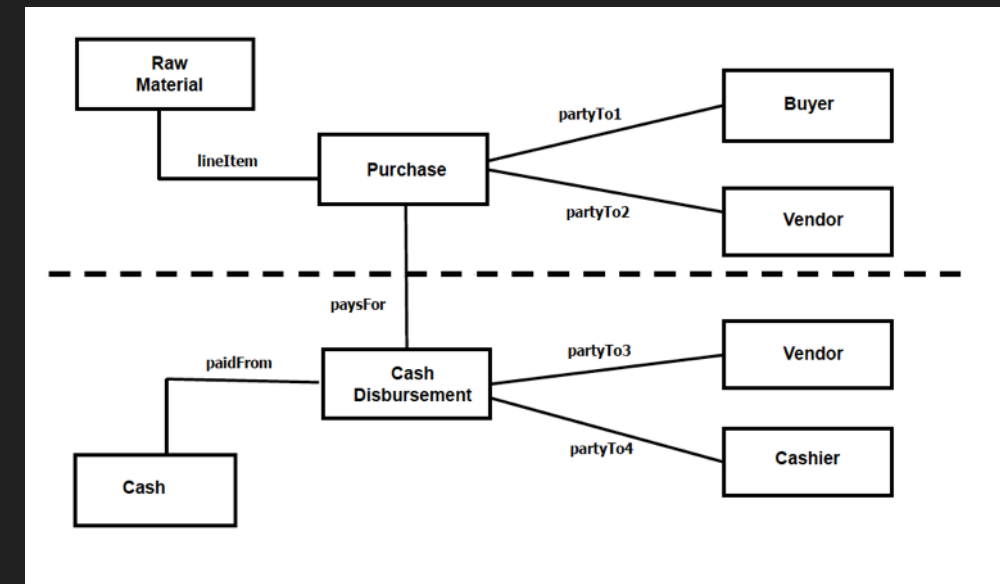
- Here's an example of the model looks applied to an actual business problem.
- The duplication of the model above and below the line reflects the concept of duality, and conveys the ideas of a debit and credit representation for the business event
- Duplicate agents typically are internal (employee) and external (customer, vendor)



Examples and Importance of Perspective Independent Definition

Event	Duality	Resource	Arrangement	Agent 1 – In	Agent 2 -Out	Commitment
Airline Ticketing	Cash Receipt	Cash	Ticket Contract	Ticket Agent	Customer	Payment
	Commit a seat	Seats	Ticket Contract	Customer	Ticket Agent	Ticket
Deposit	Cash Receipt	Cash	Checking Account	Teller	Customer	Legitimacy of the Payment
	Increase Checking Account	Checking Account	Checking Account	Customer	Teller	Availability of funds, perhaps interest, etc.

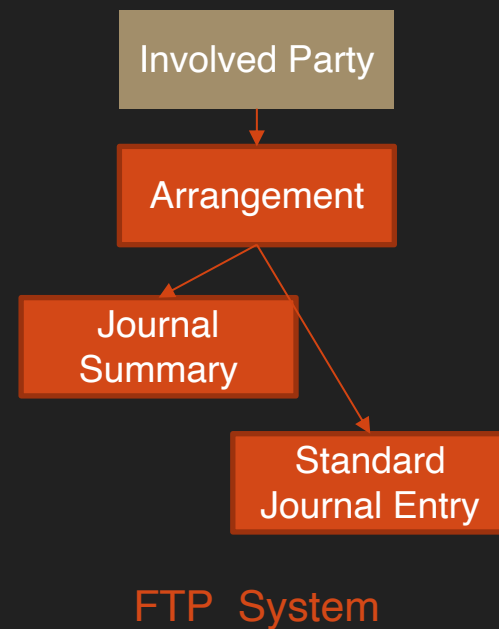
- The example to the right could be put into the table above.
- The additional examples above give a sense of how this tool is used to define fundamental building blocks of business activity.
- These building blocks are independent of perspective and reporting needs.
- The last row Deposit Event could easily commingle perspective, when it is more powerful to avoid doing so. For example:
 - If the chart were written from the Customer's perspective, the bottom line might be "Increase Asset"
 - Whereas the same row written from the Bank's perspective might be, "Increase Liability"
- Imposing perspective confuses the definition of a shared ledger. Shared ledgers must be defined independent of participating perspective.



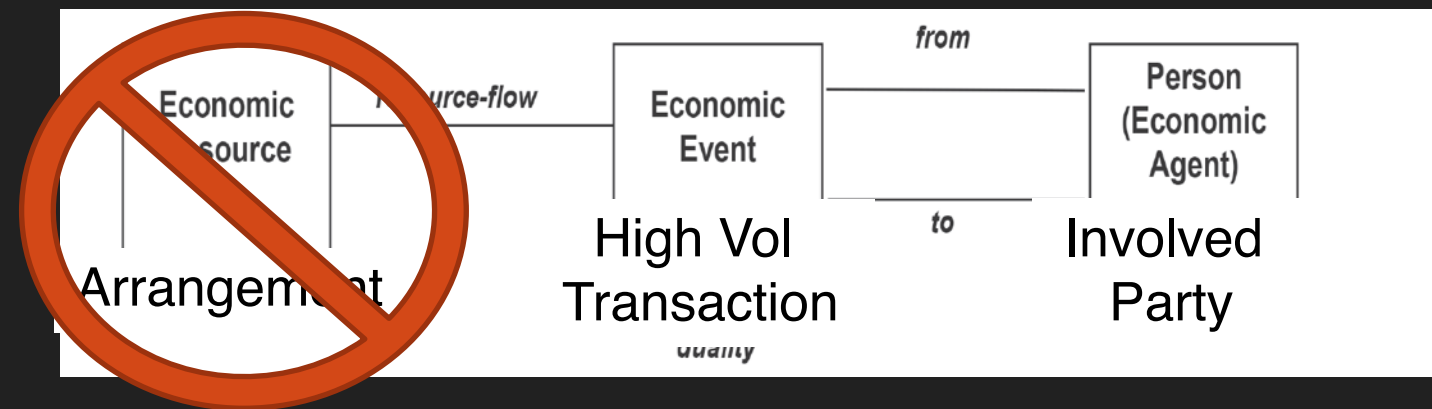
The above specific example is formed from the table below, but it does not include commitment or contract specifications

FTP to REA Relationship

- If FTP's Arrangement is really a term for Contract, then the FTP data model does NOT correspond to the historical REA entities, but instead includes another entity.
- Resource is an entity for Inventory, which is not really used in Financial Services, except for perhaps Facility.
- Considering the physical FTP implementation, the entities here in **Orange** are Common Key Buffer entities. Involved Party is accessed via a direct lookup.



Arrangement



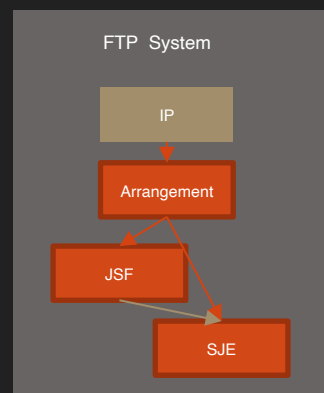
Possible Base Model

Marrying FTP and REA approaches might result in something like this data model

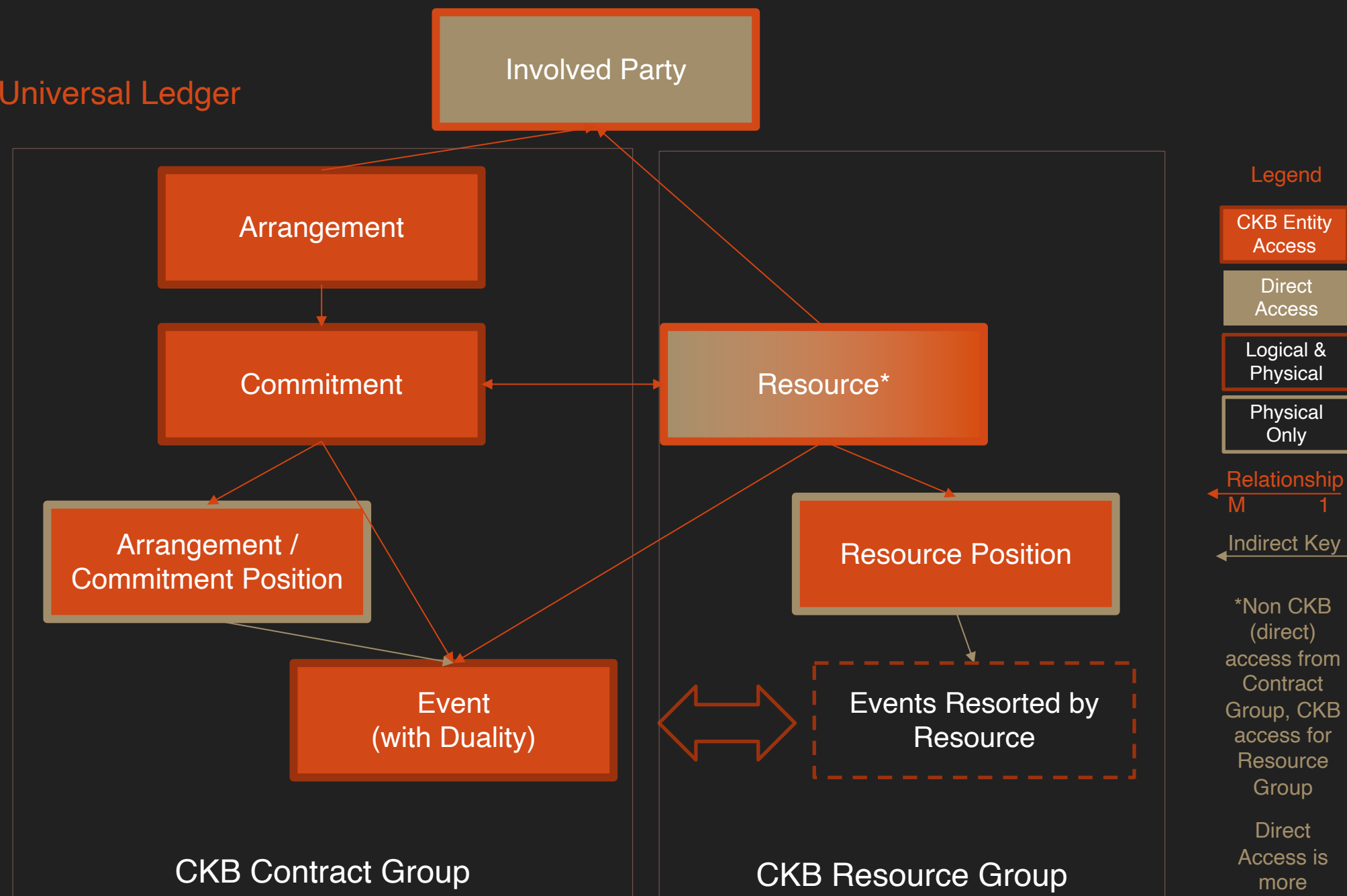
This is not simply a logical view, but also incorporates physical elements in order to provide efficient scale

Changes to FTP include:

- Rename Arrangement to Contract
- Add Commitment
- Add Resource and Resource Position
- Define CKB Groups
- Recognize need to resort events for Resource Group



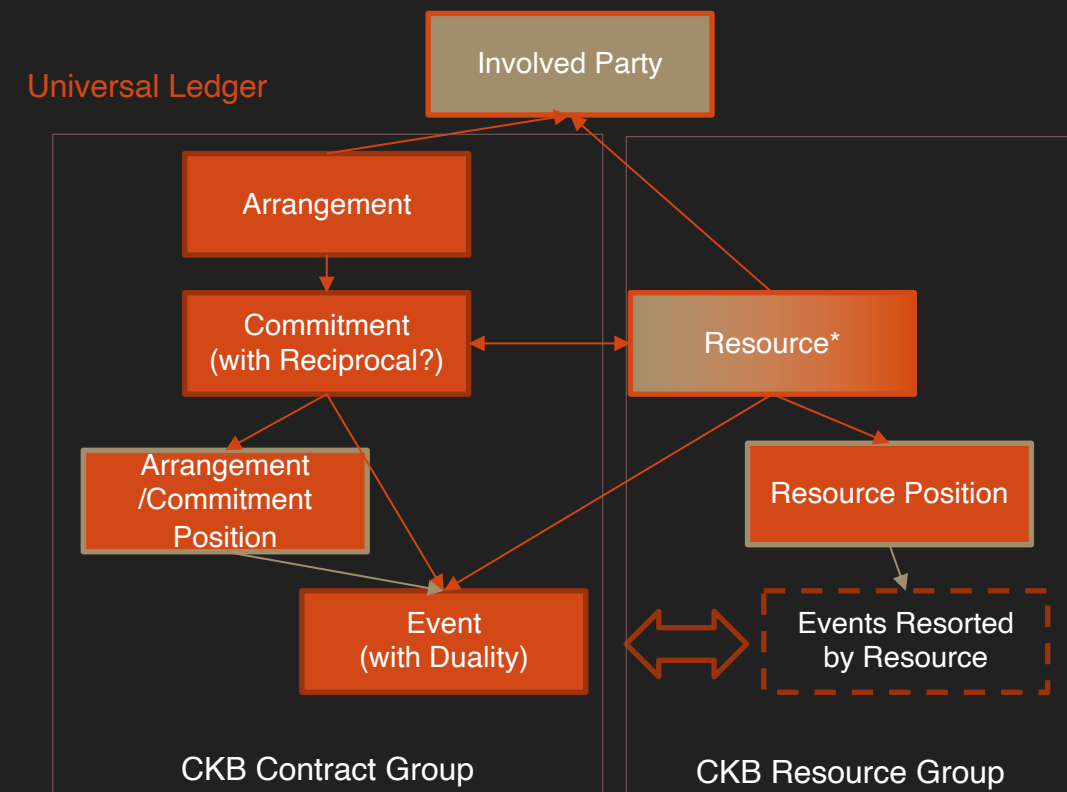
Universal Ledger



Enhanced Examples using Proposed Model

Event	Duality	Resource	Resource Position	Arrangement	Involved Parties		Commitment	Contract Position	Commitment Position
Airline Ticketing	Cash Receipt	Cash	Airline Cash Account Balance	Ticket Contract	Ticket Agent	Customer	Payment	Customer (Frequent Flyer) Sales History	Ticket Price
	Commit a seat	Seats	Aircraft Seat Inventory	Ticket Contract	Customer	Ticket Agent	Ticket	History of seats for Customer Account	Number of Seats in Ticket
Deposit	Cash Receipt	Cash	Bank Cash Account Balance	Checking Account	Teller	Customer	Legitimacy of the Payment	Bank Cash Account Balance	TBD
	Increase Checking Account	Checking Account	Customer Checking Account Balance	Checking Account	Customer	Teller	Availability of funds, perhaps interest, etc.	Customer Checking Account Balance	TBD

- The example table has been enhanced to show changes needed to reflect our proposed data model.
- Note that because banking does not typically involve tracking resources outside of the account balances associated with contracts, the Contract Position and the Resource Positions contain the same data
- The Commitment entity in financial services may provide value which for FTP was captured in the “sub arrangement” entities, which were simply part of the Arrangement entity.
- The idea of Commitment Reciprocity in the REA model needs to be researched

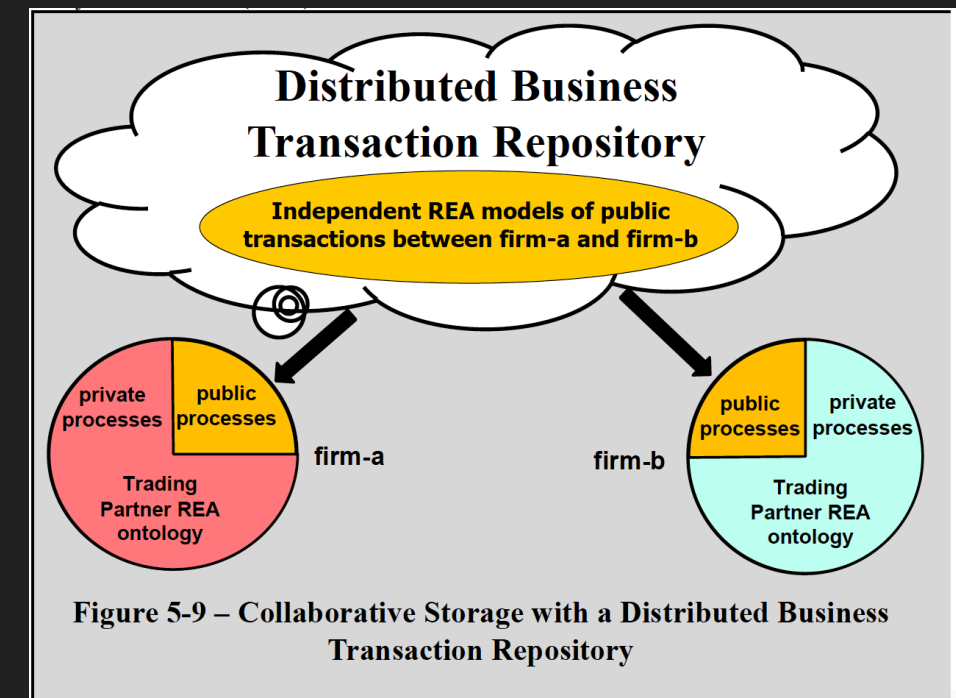


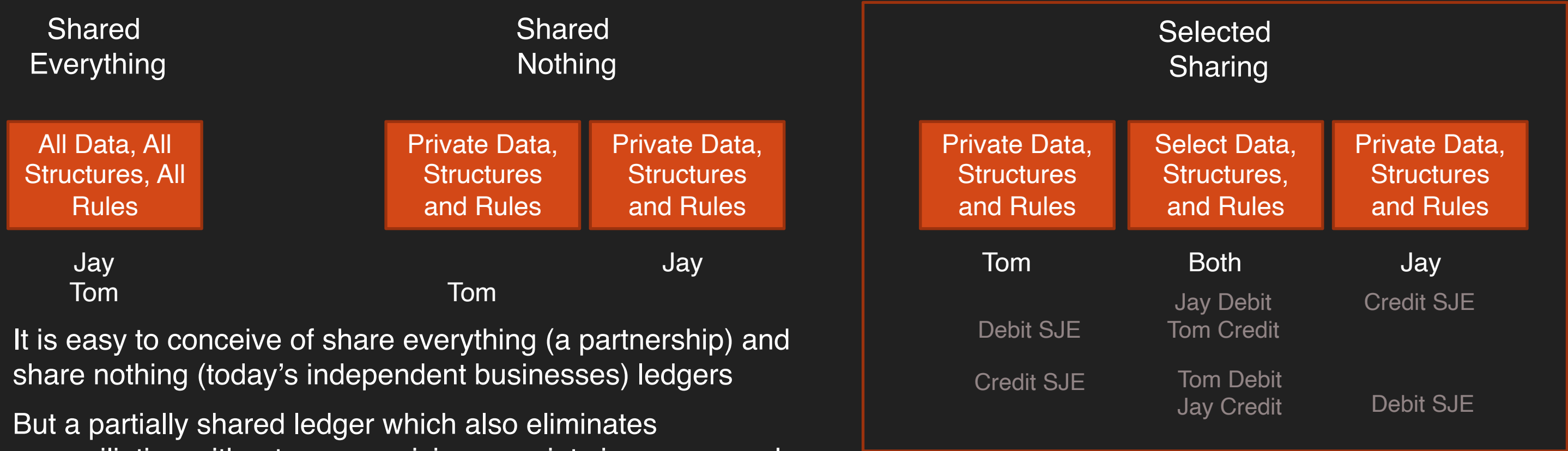
Chapter 5 of the forthcoming monograph begins to investigate potential for shared business systems. The following are quotes:

The **inside-the-firm trading partner view** that we have presented in the first four chapters will still dominate in practice, because firms will still have established boundaries whose performance metrics (including traditional financial statements) will remain important and whose business processes **will almost assuredly include a large set of internal conversion processes with information components those firms consider proprietary**. In that sense, the REA trading partner view becomes the conceptual foundation for the non-legacy ERP systems developing now in current practice. However, as we will discuss in the rest of this chapter, the real innovations to come in enterprise systems will feature the **independent view of REA, because increasingly, more independent economic agents will be using collaboration space to store their market exchange information**.

[The REA standard in] ISO 15944-4 (2007)....was **developed before the technological revolution of cryptocurrencies** was introduced by Satoshi Nakamoto (2008), but the blockchain technology ideas initiated there (and developed extensively since) have created the perfect technology platform [this might be argued] for the ideas of collaboration space and the independent view of the REA model.... The business processes in each firm's value chain would be divided into two categories:

1. **Private processes** to include all conversions plus those market exchanges not transacted in the distributed repository. These would be governed by the REA trading partner view, perhaps as implemented in an ERP system.
2. **Public processes** to include exchanges transacted in collaboration space. Those business transactions would only be recorded on the blockchain with an REA independent view of the ontological categories, but their effect would be aligned with the individual firm trading partner view via transformation rules of the type pioneered by Laurier et al. (2017).” (“The REA Accounting Model as an Accounting and Economic Ontology” pages 94 and 95)





- It is easy to conceive of share everything (a partnership) and share nothing (today’s independent businesses) ledgers
- But a partially shared ledger which also eliminates reconciliation without compromising propriety is more complex
- The selected share ledger requires agreeing on what to share for "direct" postings from transactions
- A key component to this will be definition and agreement by both parties on the rules (perhaps called accounting rules) on what the format of this transaction or one-sided journal will be
- In some cases (perhaps most?) the “duality” or accounting “offset” transaction will not be shared. In these cases, this offset will need to be held privately
- Rules for these offsets and reference data and encoding schemes will also be private

Shared Partitioning of Universal Ledger

- A higher-level Involved Party Participant entity must be created which recognizes the shared relationship between the two parties.
- This table will contain: (1) A Unique IP ID (key), (2) A list of all participating IP IDs, like Tom and Jay's
- This Unique IP ID will create another set of Universal Ledger partitions for all the shared data.
- Private, offset journal data will generally be generated from private rules reading the shared partition, and routed for storage, additional processing and reporting use to the private partitions

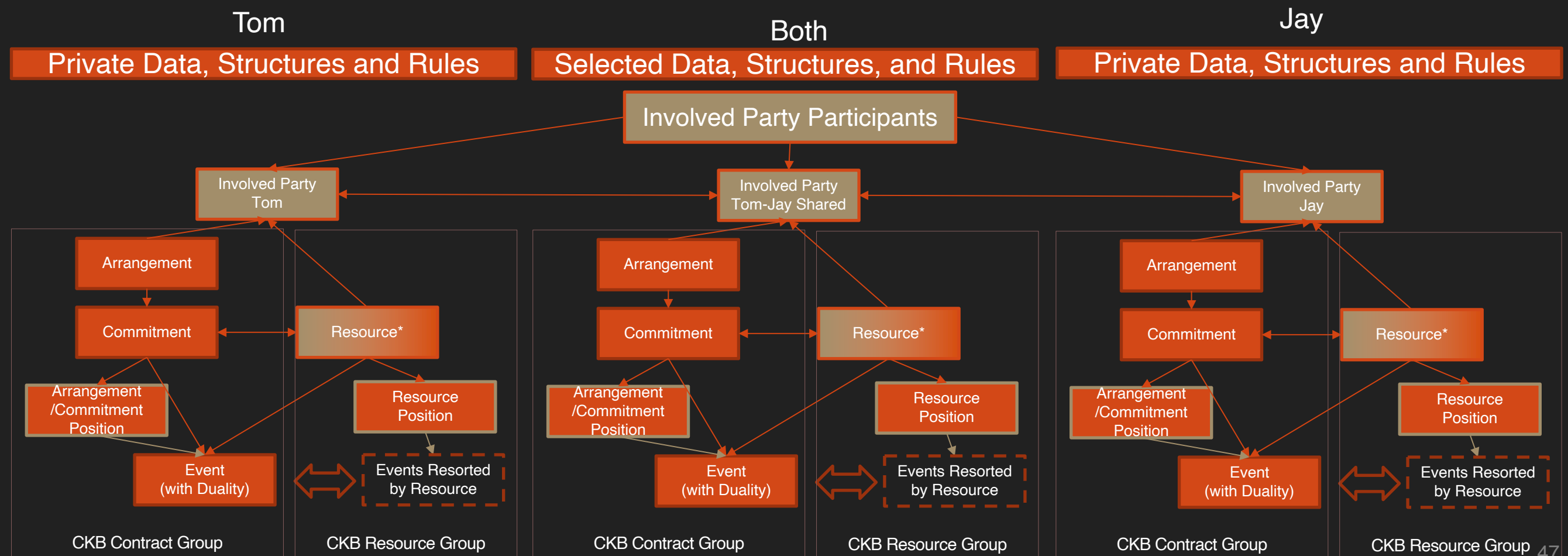




Chart of Account Setup for a typical shared transaction of:

- Vendor Tom
- Sales
- Customer Jay
- Packaged Cement
- On Account

Tom

4 Creates Private Ledger

Creates Translation from Shared Data

Shared COA	Acct	Title	Shared (Y/N)
A	1400	Cement For Sale	Y
B	1200	Accounts Receivable	Y

1 Tom Creates a Ledger

Tom Authorizes Partner Jay 2

3 Agree Shared Rules/Data

Shared COA	Tom Title	Jay Title
A	Goods for Sale	Raw Materials
B	AR	AP

Jay

4 Creates Private Ledger

Creates Translation from Shared Data

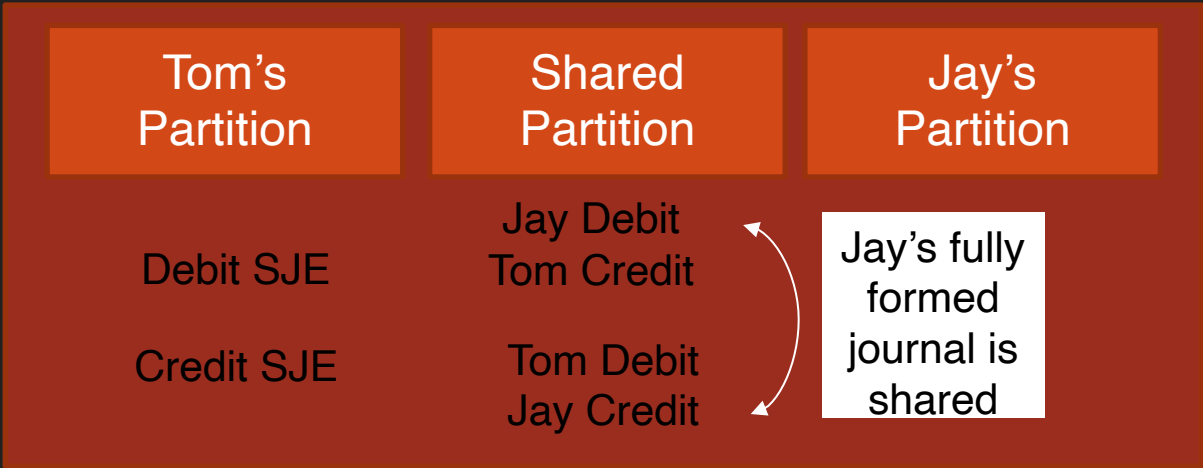
Shared COA	Acct	Title	Shared (Y/N)
A	13-2	Raw Material - Cement	Y
B	25-7	Accounts Payable	Y

Sharing a ledger likely starts by:

1. One party creating a ledger (Tom)
2. Inviting the counterpart(ies) to participate in the ledger
3. Agreeing on what the values are in the shared records, and what the rules are to create them
4. Each party then creates their own interpretation of the shared data for their own perspectives

This last point is necessary because:

- The parties to the transactions will likely not want to base all their accounting on shared reference data. For example, if the shared transactions included a shared chart of accounts, both parties would have to agree on all elements of the chart.
- Doing so would also preclude differing perspectives on the same business event. In other words, Tom and Jay would both show the same transaction as a debit and credit, not opposite.
- In this example, the Shared Chart of Account (COA) value is A and B, but for Tom, A is Goods for Sale (inventory), whereas for Jay it is a Raw Material.



- The duality of transactions (debit and credit) may be visible in some cases in the shared partitions.
- In this example, both sides of Jay's entries are stored in the shared partition.
- However, Tom's perspectives requires:
 - Additional reference data entries which are not shared
 - Offsets stored in his private partition
- This may happen because:
 - Tom's perspective has Income Statement impacts
 - But Jay's does not, only affecting the balance sheet

Tom

Shared COA	Acct	Title	Shared (Y/N)
A	1400	Cement For Sale (Inv)	Y
B	1200	Accounts Receivable	Y
	5200	Cost of Goods Sold	N
	4100	Sales Revenue	N

Jay

Shared COA	Acct	Title	Shared (Y/N)
A	13-2	Raw Material - Cement	Y
B	25-7	Accounts Payable	Y

- This example may need to be refined because the Cost of Good's Sold dollar amount is typically different from the Revenue amount and would not be shared with the counterparty.
- Additional work is needed to test the types of conditions that would be created.
- Also, it may be necessary to suspend the FPEM edit requiring a balanced set of entries in the individual partitions.
- The edit should be applied across each participant's full set of books considered in total, both the shared and the private partitions.

Conclusion: My Predictions

April 15, 2020

- Any currency is fundamentally transferrable credit, whether issued by a government or any other entity. The most effective type of transferable credit over time is that which is backed by integrity--the commitment of the person issuing the promissory note to repay--as the value of any other asset in the future will always be uncertain.
- Trustworthiness begets efficiencies, efficiencies not available in any other way. Its power is akin to that of intelligence and knowledge. Those who are trustworthy will continue to find ways to enhance the efficiencies from interacting with those also committed to trustworthiness and integrity.
- In the same way fundamental bookkeeping increases intelligence by accurately and understandably recording what has happened and projecting what will happen, trustworthiness and integrity increase efficiencies through community, sharing, specialization and mutual dependence. "Trustlessness," one of the Bitcoin blockchain goals, is a dead-end road.
- The forgotten knowledge embedded in our financial systems, including what a ledger is and how to improve it, can be recovered and enhanced, and the incredible efficiencies created by those who have gone before can be increased, and extended to perhaps billions of people, including the most simple and poorest merchants in the world, enabling access to knowledge through bookkeeping, at reduced costs through sharing, distributed by low-cost technologies.
- The resulting system will fundamentally be about organized, accurate, updatable, understandable, secure and accessible financial data. It will be broadly controlled by the owners of the data, their partners, both directly and speaking through their governments and other institutions. The change may be so radical it may upend the current concentration of capital.
- These changes will be abetted by the failure of trustworthiness and integrity in some existing financial and business institutions, and furthered by the courage, intelligence and caring of the many committed to integrity.
- I believe I will live to see a good portion of these predictions realized, in one way or another.



Appendix

Notes to Computing Cost Slide

- **Computing Costs Data:** This analysis uses computer disk costs as surrogate for total computing costs. Data was obtained from Matthew Komorowski's website: "A history of storage cost article updated September 8, 2009 (2014 update)" Accessed May 22, 2018: <http://www.mkomo.com/cost-per-gigabyte> Komoroski's sources are not public, but this series of data is used in many such analyses. His sample of disk costs from 1980 to 2014 shows that costs decline based upon this algorithm: $=\text{POWER}(10, -0.2502 * (\text{year} - 1980) + 6.304)$ with an r squared correlation of 0.9916. I selected the cost for the first entry in his table for each year (rather than an average).
- To extend this data to 1964, I used IBM press releases from the IBM Archivist from 1964, 1966, 1969. I did not adjust these numbers for inflation. Additional years were reviewed for 1968, 1972, and 1976, but the cost seemed unusually high, and likely included CPU and other system costs which could not be eliminated from the available sources.
- I also reviewed Komoroski's 1980 and 1987 numbers, noting that the IBM disk cost for 1980 was \$173,626 per gigabyte of storage, and average for 1987 was \$106,135, compared to Komoroski's cost of \$193,000 and \$90,000 respectively. These comparisons give some sense of validity to his underlying data, and a reasonableness for the trend.
- Note that years before 1980 are not predicted by Komoroski's algorithm which estimated a costs of \$20 million in 1964 rather than \$6.6 million. Although Moore's law was proposed in 1965, the impact upon disk costs may have been slower in starting.
- Slide 1:
- **Technology Spending Data:** With assistance from Edward T. Morgan, Chief, Industry Sector Division, Bureau of Economic Analysis, U.S. Department of Commerce, from US Federal Bureau of Economic Analysis GDP "Value Added by Industry" Release Date: April 19, 2018, specifically the Addenda Row for the years 1997 - 2017 "Information-communications-technology-producing industries: Consists of computer and electronic product manufacturing (excluding navigational, measuring, electromedical, and control instruments manufacturing); software publishers; broadcasting and telecommunications; data processing, hosting and related services; internet publishing and broadcasting and web search portals; and computer systems design and related services. Accessed May 22, 2018
<https://bea.gov/iTable/iTable.cfm?reqid=51&step=51&isuri=1&5114=a&5102=1#reqid=51&step=51&isuri=1&5101=1&5114=a&5113=22r&5112=1&5111=1997&5102=1>
- The Technology Addenda data is not calculated before 1997; to estimate data prior to 1997 I used the same combination of rows used to calculate the addenda data based upon row titles value and addenda description, from the GDP series data https://www.bea.gov/industry/xls/io-annual/GDPbyInd_VA_1947-2017.xlsx. These values were tested against IBM total revenue numbers from IBM annual reports for each year, provided by IBM Archivist. The values for "Publishing industries, except internet (includes software)" were excluded for years before 1990 based upon the IBM revenue numbers.